MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DRIVER, BUS, DIGITAL, KEY CODE GQB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic bus driver module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.108 failures per million operating hours for class I or 0.291 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GQB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/266-1 for class I and M28787/266-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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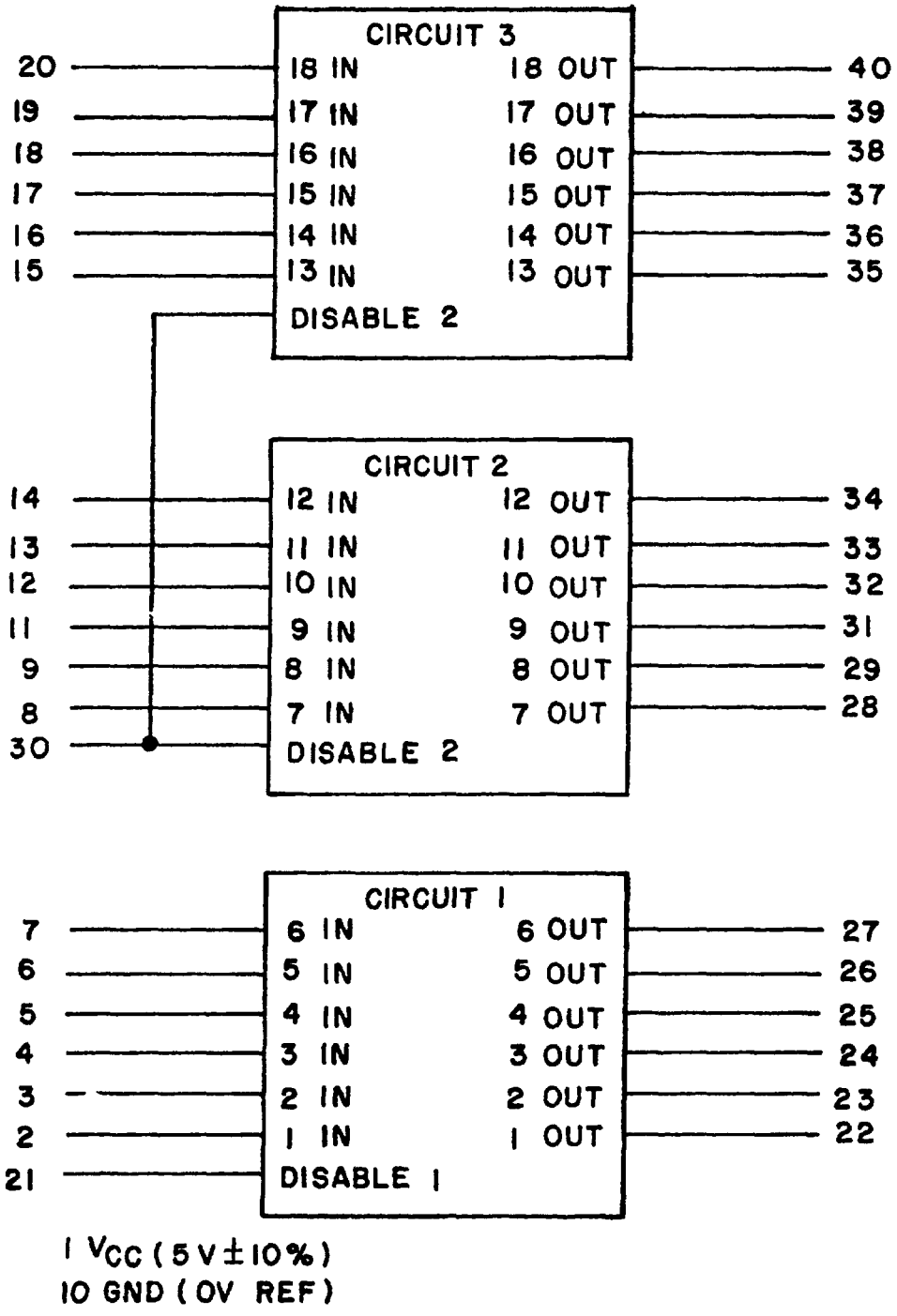


FIGURE 1. GQB module function diagram, bus driver.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C19	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
C20	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
D1-D18	Output short circuit current	VI	X	-40	-130	-40	-130	-40	-130	-40	-130	-40	-130	mA
E1-E20	Input clamp diode voltage	VII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G18	Output leakage current, high level	VIII	X	-40	40	-40	40	-40	40	-40	40	-40	40	µA
H1-H18	Output leakage current, power off	IX	X	0	40	0	40	0	40	0	40	0	40	µA
J1-J18	Output leakage current, low level	X	X	-40	40	-40	40	-40	40	-40	40	-40	40	µA
K1-K18	Output voltage, low level (I_{OL} = 32 mA)	XI	X	0	500	0	500	0	500	0	500	0	500	mV
L1-L18	Output voltage, high level	XII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
M1-M19	Input current, high level	XIII	X	0	40	0	40	0	40	0	40	0	40	µA
M20	Input current, high level	XIII	X	0	80	0	80	0	80	0	80	0	80	µA
N1-N19	Input current, low level	XIV	X	0	-1.76	0	-1.76	0	-1.76	0	-1.76	0	-1.76	mA

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N20	Input current, low level	XIV	X	0	-3.52	0	-3.52	0	-3.52	0	-3.52	0	-3.52	mA
N21-N38	Input current, low level	XIV	X	0	-1.76	0	-1.76	0	-1.76	0	-1.76	0	-1.76	mA
P	Power supply current	XV	X	60	294	60	294	60	294	60	294	60	294	mA
Q1-Q18	Transition time (t_{TLH}), C_L = 50 pF	XVI	S	4	40	4	40	4	45	4	40	4	48	ns
R1-R18	Transition time (t_{THL}), C_L = 50 pF	XVI	S	2	9	2	9	2	11	2	9	2	12	ns
S1-S18	Propagation delay time (t_{PLH}), C_L = 50 pF	XVII	S	3	16	3	16	3	19	3	16	3	20	ns
S19-S36	Propagation delay time (t_{PLH}), C_L = 1,000 pF	XVII	Q	8	63	8	63	8	68	8	63	8	71	ns
T1-T18	Propagation delay time (t_{PHL}), C_L = 50 pF	XVII	S	3	22	3	22	3	25	3	22	3	26	ns
T19-T36	Propagation delay time (t_{PHL}), C_L = 1,000 pF	XVII	Q	8	70	8	70	8	75	8	70	8	81	ns
U1-U18	Disable delay time (t_{oh}), $C_L \leq 10$ pF	XVIII	S	2	27	2	27	2	32	2	27	2	35	ns
V1-V18	Disable delay time (t_{ho}), C_L = 50 pF	XVIII	S	2	37	2	37	2	42	2	37	2	45	ns
W1-W18	Disable delay time (t_{lh}), $C_L \leq 10$ pF	XIX	S	2	16	2	16	2	21	2	16	2	24	ns
X1-X18	Disable delay time (t_{hl}), C_L = 50 pF	XIX	S	2	40	2	40	2	45	2	45	2	48	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y18	Output crosstalk, high level	XX	Q	0	700	0	700	0	700	0	700	0	700	mV
Z1-Z18	Output crosstalk, low level	XXI	Q	0	1500	0	1500	0	1500	0	1500	0	1500	mV
AA	Power supply transient amplitude	XXII	Q	0	400	0	400	0	400	0	400	0	400	mV
BB	Power supply filter capacitance	XXIII	S	0.77	1.43	0.77	1.43	0.77	1.43	0.77	1.43	0.77	1.43	µF
CC1-CC19	Input capacitance	XXIV	Q	2	15	2	15	2	15	2	15	2	15	pF
CC20	Input capacitance	XXIV	Q	4	25	4	25	4	25	4	25	4	25	pF
DD1-DD18	Output capacitance	XXV	Q	4	25	4	25	4	25	4	25	4	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through DD shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through DD in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through BB shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition BB, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through DD shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

		Class I	Class II	Unit
Supply voltage		5 ±10 percent	5 ±10 percent	V dc
Maximum power dissipation (outputs not loaded)		1.62	1.62	W
Input voltage:				
Low level		-0.3 min, 0.8 max	-0.3 min, 0.8 max	V
High level		2 min, V_{CC} max	2 min, V_{CC} max	V
Input current, disable low:				
Low level	(V_{IL} = 0.0 V)	0 min, -1.76 max	0 min, -1.76 max	mA
	(V_{IL} = 0.0 V), disable 2	0 min, -3.52 max	0 min, -3.52 max	mA
Input current, disable high:				
Low level	(V_{IL} = 0.0 V)	0 min, -1.76 max	0 min, -1.76 max	mA
High level	(V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
	(V_{IH} = 5.5 V)	0 min, 1 max	0 min, 1 max	mA
	(V_{IH} = 2.4 V), disable 2	0 min, 80 max	0 min, 80 max	µA
	(V_{IH} = 5.5 V), disable 2	0 min, 2 max	0 min, 2 max	mA
Output voltage				
Low level	(I_{OL} = 32 mA)	0.0 min, 0.5 max	0.0 min, 0.5 max	V
High level	(I_{OH} = -2 mA)	2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
Output drive current				
Low level	(V_{OL} = 0.5 V)	32 max	32 max	mA
High level	(V_{OH} = 2.4 V)	-2 max	-2 max	mA
Output short circuit current		-40 min, -130 max	-40 min, -130 max	mA
Output leakage current				
V_{OL} = 0.5 V		-40 min, 40 max	-40 min, 40 max	µA
V_{OH} = 2.4 V		-40 min, 40 max	-40 min, 40 max	µA
Propagation delay time (C_L = 1,000 pF)				
Input to output (t_{PLH})		8 min, 65 max	8 min, 71 max	ns
Input to output (t_{PHL})		8 min, 70 max	8 min, 81 max	ns
Propagation delay time (C_L = 50 pF)				
Input to output (t_{PLH})		3 min, 16 max	3 min, 20 max	ns
Input to output (t_{PHL})		3 min, 22 max	3 min, 26 max	ns
Disable to output (T_{1h}) (C_L ($\leq$ 10 pF)		2 min, 16 max	2 min, 26 max	ns
Disable to output (T_{oh}) (C_L ($\leq$ 10 pF)		2 min, 27 max	2 min, 35 max	ns
Disable to output (T_{ho})		2 min, 37 max	2 min, 45 max	ns
Disable to output (T_{h1})		2 min, 40 max	2 min, 48 max	ns

* 6.5 Module function. This module contains three 6-input three-state line drivers. Each driver operates in a single ended driver mode or a high-impedance output mode.

6.5.1 Single-ended driver mode. To operate as 18 single-ended line drivers, apply low level to the output disable inputs (to keep the outputs in the normal low-impedance mode). All channels operate with no signal inversion.

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6.5.2 High-impedance output mode. To operate in a bus-oriented system with outputs tied directly to outputs of similar devices, all devices except one must be placed in the high-impedance state. This is accomplished by insuring that a high-level signal is applied to the output disable inputs of all devices except the one that is used to drive the bus. The driving device provides good capacitive drive capability and waveform integrity, especially on the positive transition.

6.5.3 Input to output relationship. Each driver has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Inputs		Outputs
Disable	In	Out
L	L	L
L	H	H
H	X	High-impedance state

1/ H = High level.
2/ L = Low level.
3/ X = Don't care.

6.5.4 Parallel outputs. Paralleling of outputs is permissible provided only one output at a time is in a high or low state and all other outputs are in the high-impedance state.

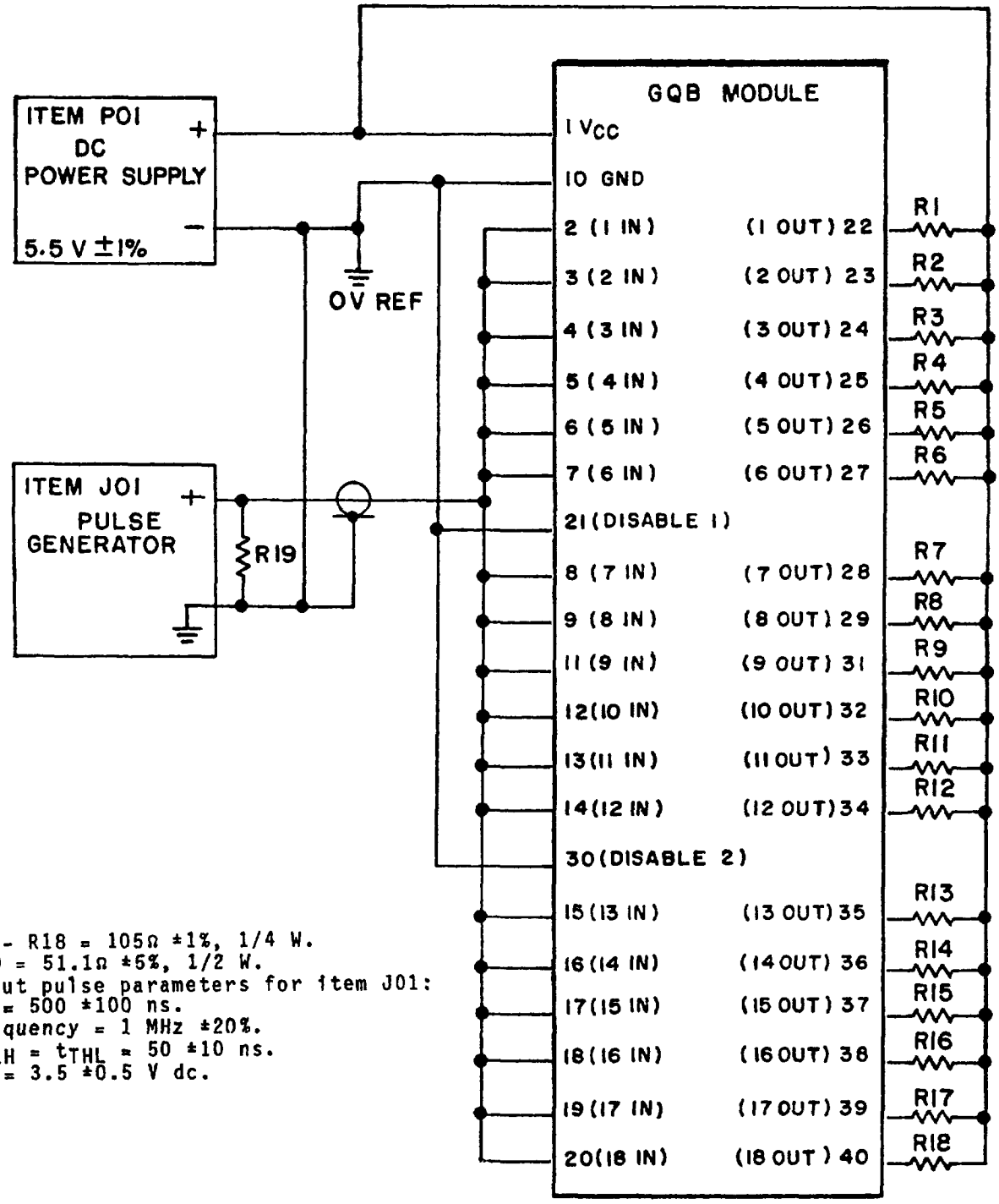
6.6 Changes for previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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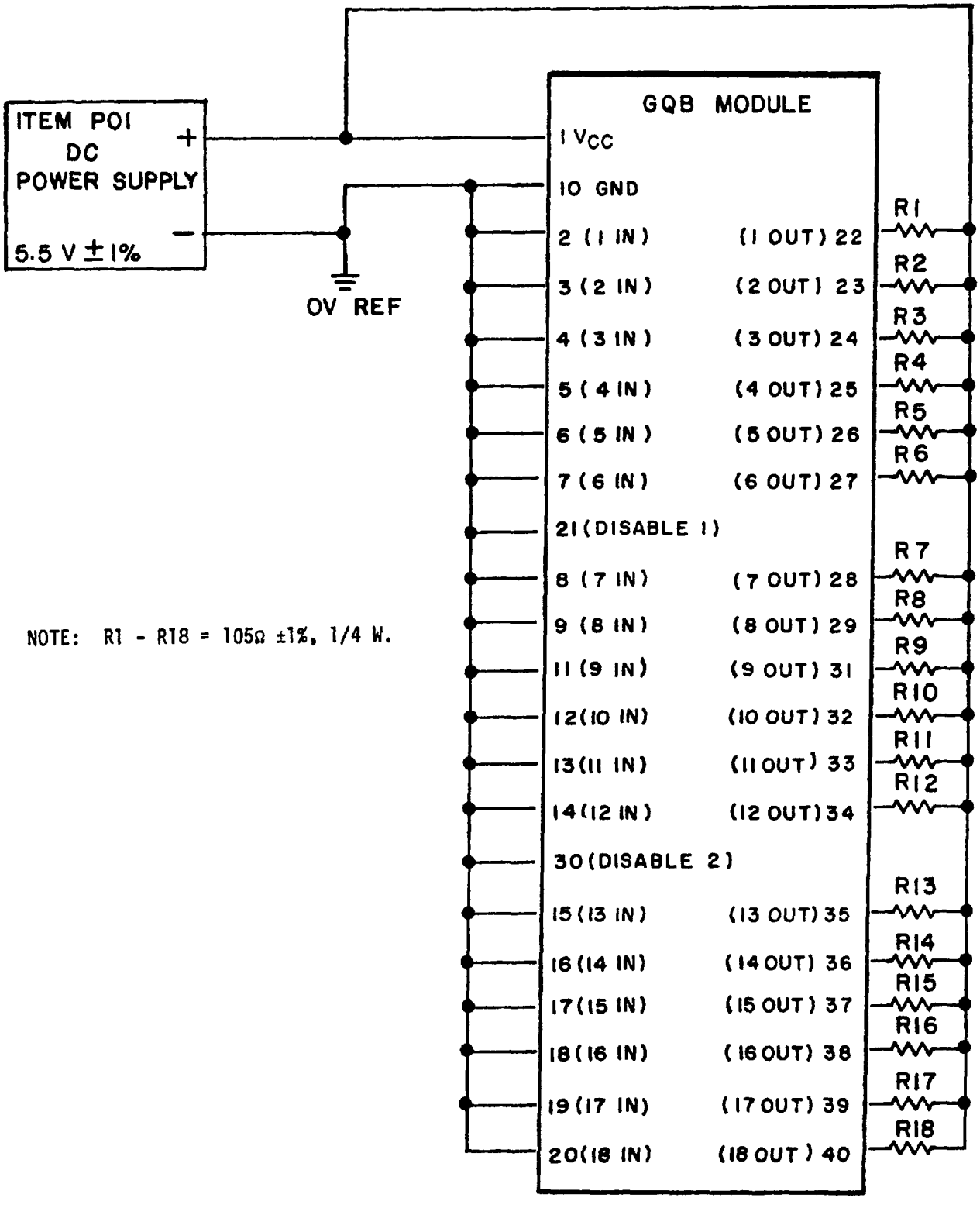
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NOTES:
1. R1 - R18 = 105Ω ±1%, 1/4 W.
2. R19 = 51.1Ω ±5%, 1/2 W.
3. Input pulse parameters for item J01:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL} = 50 \pm 10$ ns.
   PA = 3.5 ±0.5 V dc.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1 - R18 = 105Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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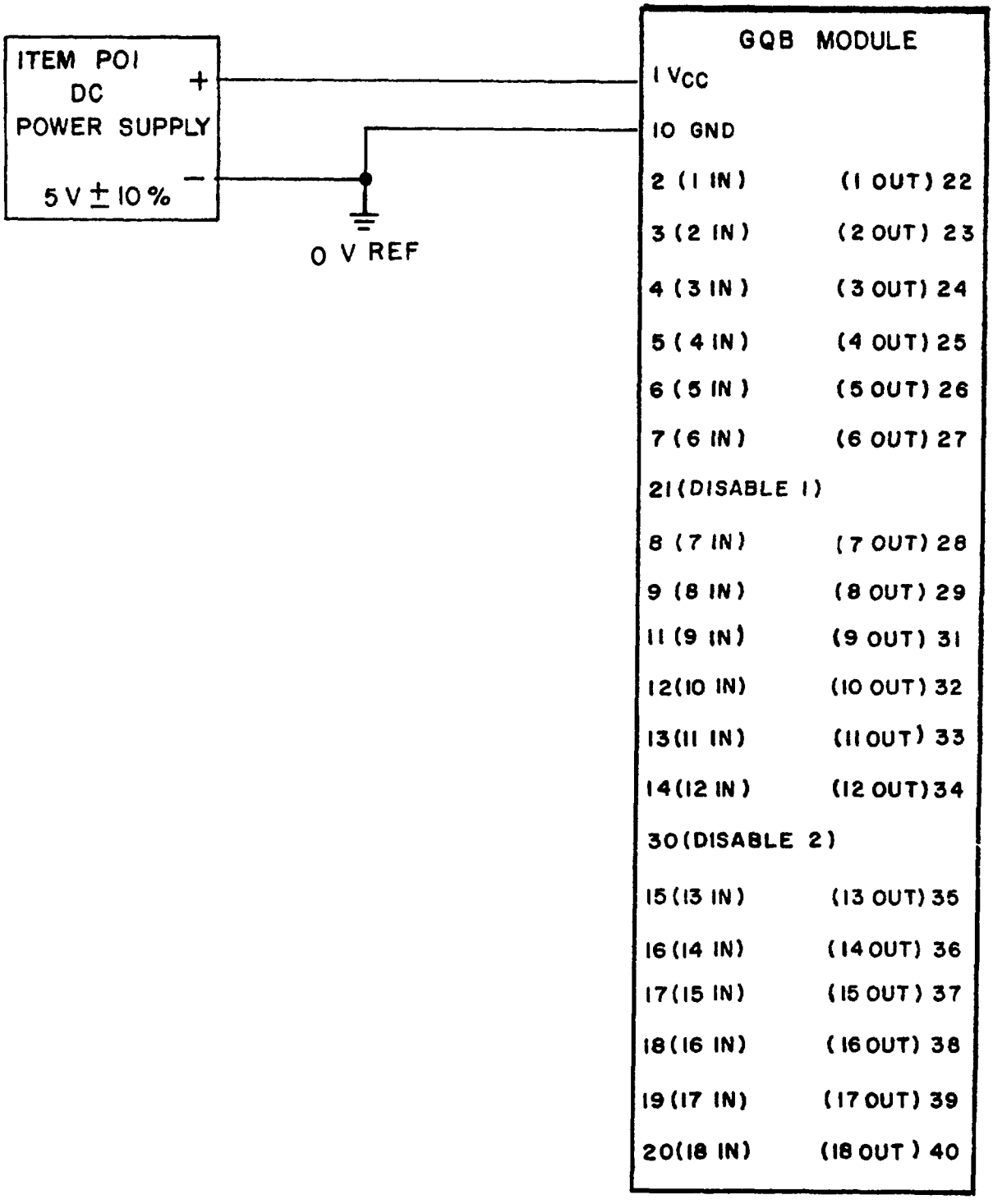


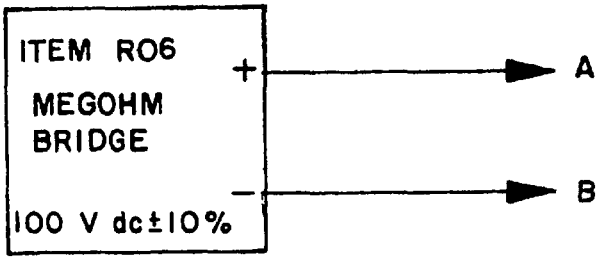
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Module frame	10	∞	MΩ	5

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

GQB MODULE	
1 V_{CC}	
10 GND	
2 (1 IN)	(1 OUT) 22
3 (2 IN)	(2 OUT) 23
4 (3 IN)	(3 OUT) 24
5 (4 IN)	(4 OUT) 25
6 (5 IN)	(5 OUT) 26
7 (6 IN)	(6 OUT) 27
21 (DISABLE 1)	
8 (7 IN)	(7 OUT) 28
9 (8 IN)	(8 OUT) 29
11 (9 IN)	(9 OUT) 31
12 (10 IN)	(10 OUT) 32
13 (11 IN)	(11 OUT) 33
14 (12 IN)	(12 OUT) 34
30 (DISABLE 2)	
15 (13 IN)	(13 OUT) 35
16 (14 IN)	(14 OUT) 36
17 (15 IN)	(15 OUT) 37
18 (16 IN)	(16 OUT) 38
19 (17 IN)	(17 OUT) 39
20 (18 IN)	(18 OUT) 40

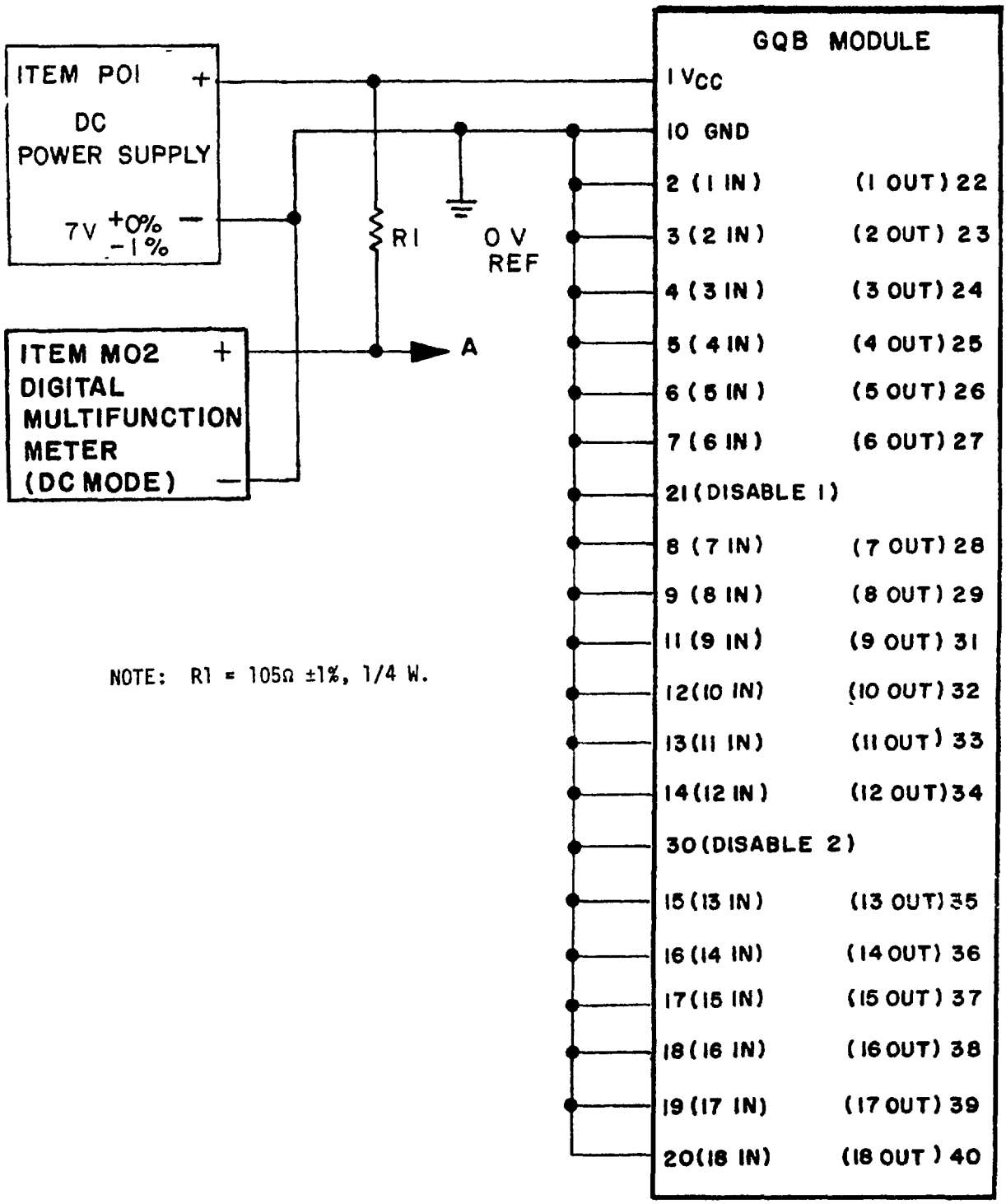
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
B1	1 OUT	V_{OUT} (low loaded)	22	0.01	1.00	V	6
B2	2 OUT	"	23	"	"	"	"
B3	3 OUT	"	24	"	"	"	"
B4	4 OUT	"	25	"	"	"	"
B5	5 OUT	"	26	"	"	"	"
B6	6 OUT	"	27	"	"	"	"
B7	7 OUT	"	28	"	"	"	"
B8	8 OUT	"	29	"	"	"	"
B9	9 OUT	"	31	"	"	"	"
B10	10 OUT	"	32	"	"	"	"
B11	11 OUT	"	33	"	"	"	"
B12	12 OUT	"	34	"	"	"	"
B13	13 OUT	"	35	"	"	"	"
B14	14 OUT	"	36	"	"	"	"
B15	15 OUT	"	37	"	"	"	"
B16	16 OUT	"	38	"	"	"	"
B17	17 OUT	"	39	"	"	"	"
B18	18 OUT	"	40	"	"	"	"

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NOTE: R1 = 105Ω ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1 IN Input breakdown current	2	Group A 1/	0	1	mA	7
C2	2 IN "	3	"	"	"	"	"
C3	3 IN "	4	"	"	"	"	"
C4	4 IN "	5	"	"	"	"	"
C5	5 IN "	6	"	"	"	"	"
C6	6 IN "	7	"	"	"	"	"
C7	DISABLE 1 "	21	"	"	"	"	"
C8	7 IN "	8	"	"	"	"	"
C9	8 IN "	9	"	"	"	"	"
C10	9 IN "	11	"	"	"	"	"
C11	10 IN "	12	"	"	"	"	"
C12	11 IN "	13	"	"	"	"	"
C13	12 IN "	14	"	"	"	"	"
C14	13 IN "	15	"	"	"	"	"
C15	14 IN "	16	"	"	"	"	"
C16	15 IN "	17	"	"	"	"	"

See footnote at end of table.

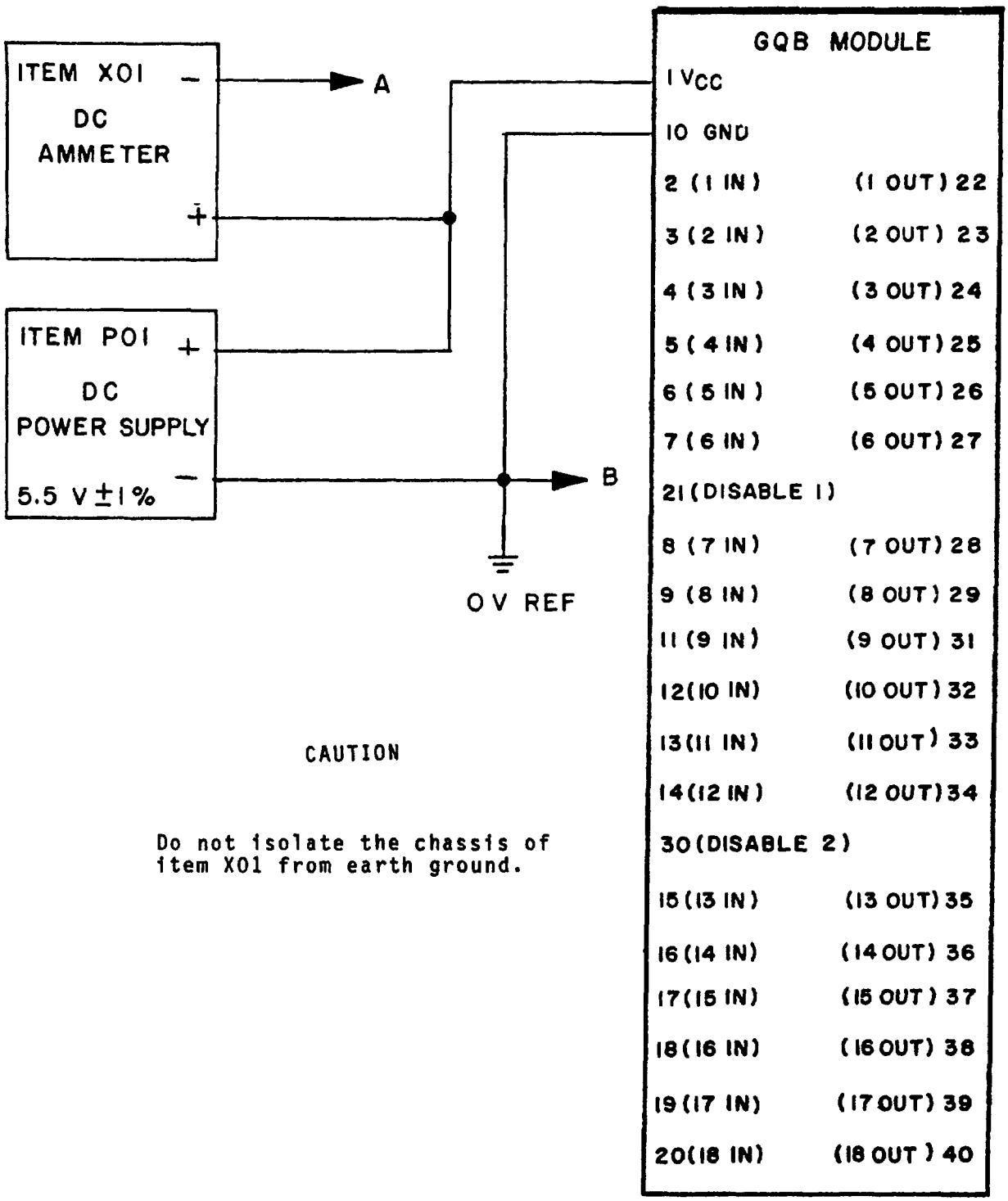
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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C17	16 IN Input breakdown current	18	Group A 1/	0	1	mA	7
C18	17 IN "	19	"	"	1	"	"
C19	18 IN "	20	"	"	1	"	"
C20	DISABLE 2 "	30	"	"	2	"	"

1/ Group A includes pins 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, and 30 except when connected to point A.

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
D1	1 OUT	Output short circuit current	2	22	-40	-130	mA	8
D2	2 OUT	"	3	23	"	"	"	"
D3	3 OUT	"	4	24	"	"	"	"
D4	4 OUT	"	5	25	"	"	"	"
D5	5 OUT	"	6	26	"	"	"	"
D6	6 OUT	"	7	27	"	"	"	"
D7	7 OUT	"	8	28	"	"	"	"
D8	8 OUT	"	9	29	"	"	"	"
D9	9 OUT	"	11	31	"	"	"	"
D10	10 OUT	"	12	32	"	"	"	"
D11	11 OUT	"	13	33	"	"	"	"
D12	12 OUT	"	14	34	"	"	"	"
D13	13 OUT	"	15	35	"	"	"	"
D14	14 OUT	"	16	36	"	"	"	"
D15	15 OUT	"	17	37	"	"	"	"
D16	16 OUT	"	18	38	"	"	"	"
D17	17 OUT	"	19	39	"	"	"	"
D18	18 OUT	"	20	40	"	"	"	"

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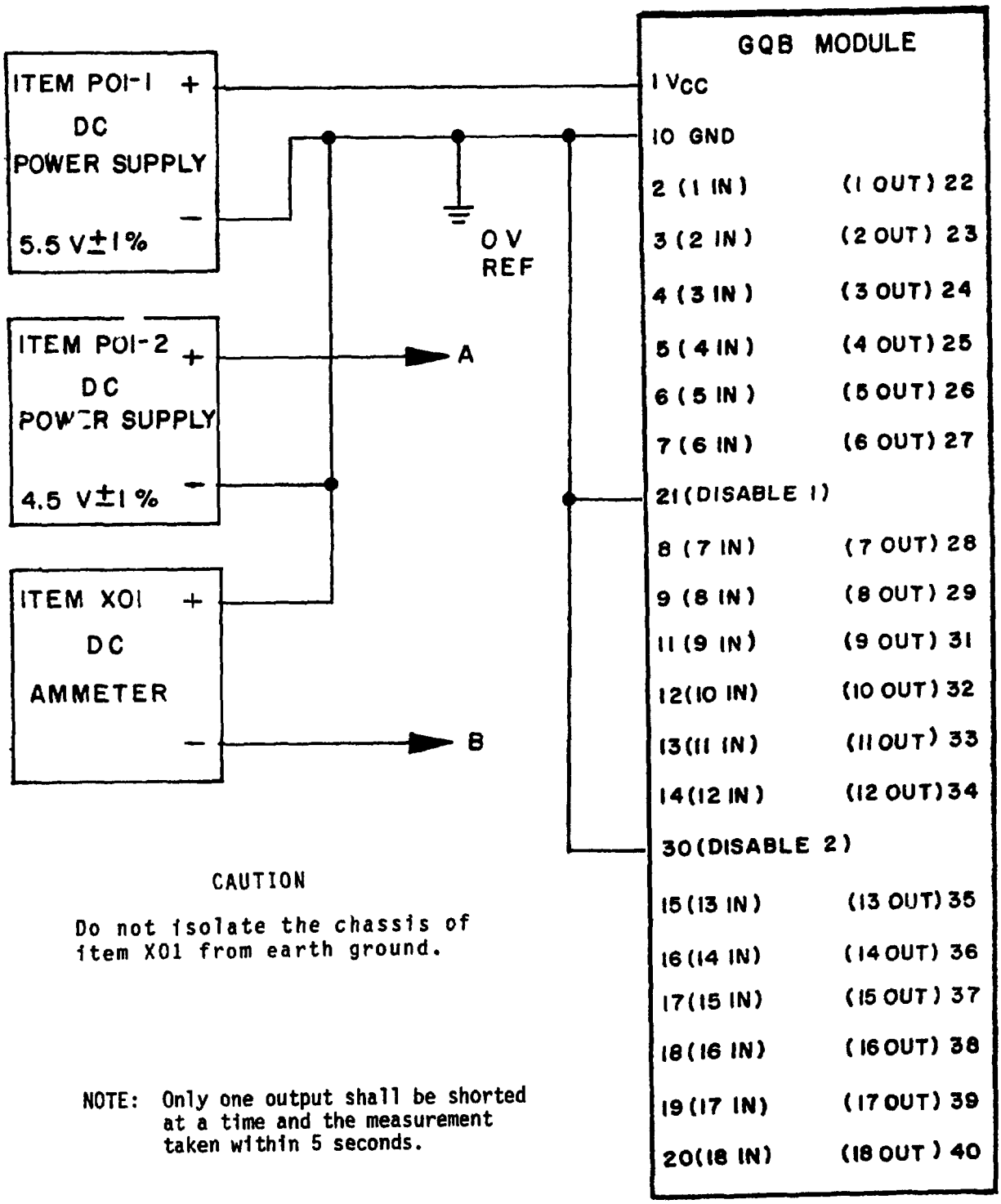


FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

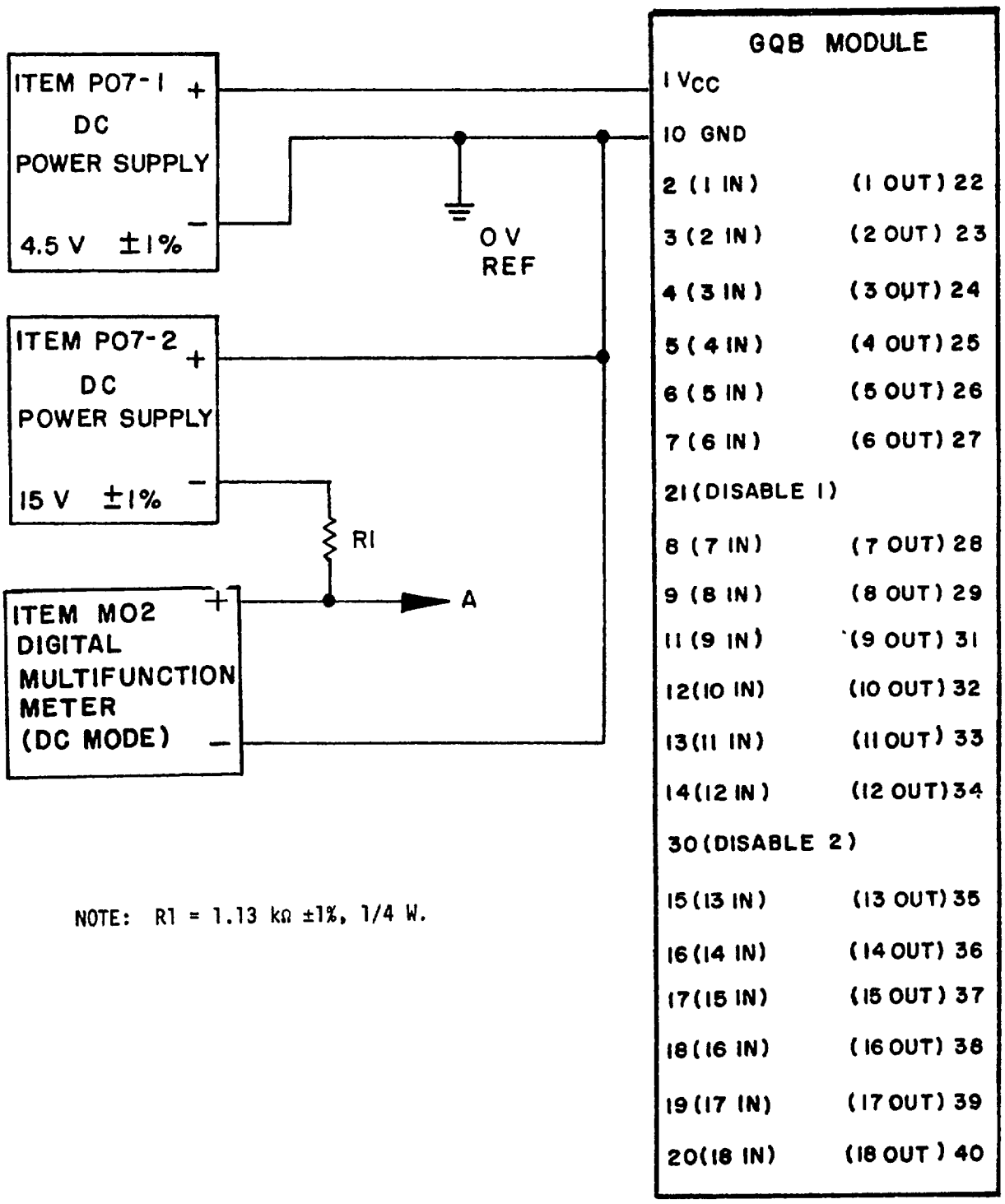
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1 IN Input clamp diode voltage	2	-0.4	-1.5	V	9
E2	2 IN "	3	"	"	"	"
E3	3 IN "	4	"	"	"	"
E4	4 IN "	5	"	"	"	"
E5	5 IN "	6	"	"	"	"
E6	6 IN "	7	"	"	"	"
E7	Disable 1 "	21	"	"	"	"
E8	7 IN "	8	"	"	"	"
E9	8 IN "	9	"	"	"	"
E10	9 IN "	11	"	"	"	"
E11	10 IN "	12	"	"	"	"
E12	11 IN "	13	"	"	"	"
E13	12 IN "	14	"	"	"	"
E14	13 IN "	15	"	"	"	"
E15	14 IN "	16	"	"	"	"
E16	15 IN "	17	"	"	"	"
E17	16 IN "	18	"	"	"	"
E18	17 IN "	19	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E19	18 IN	Input clamp diode voltage	20	-0.4	-1.5	V	9
E20	DISABLE 2	Input clamp diode voltage	30	-0.4	-1.5	V	9

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NOTE: R1 = 1.13 kΩ ±1%, 1/4 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Output leakage current (high-level) test.

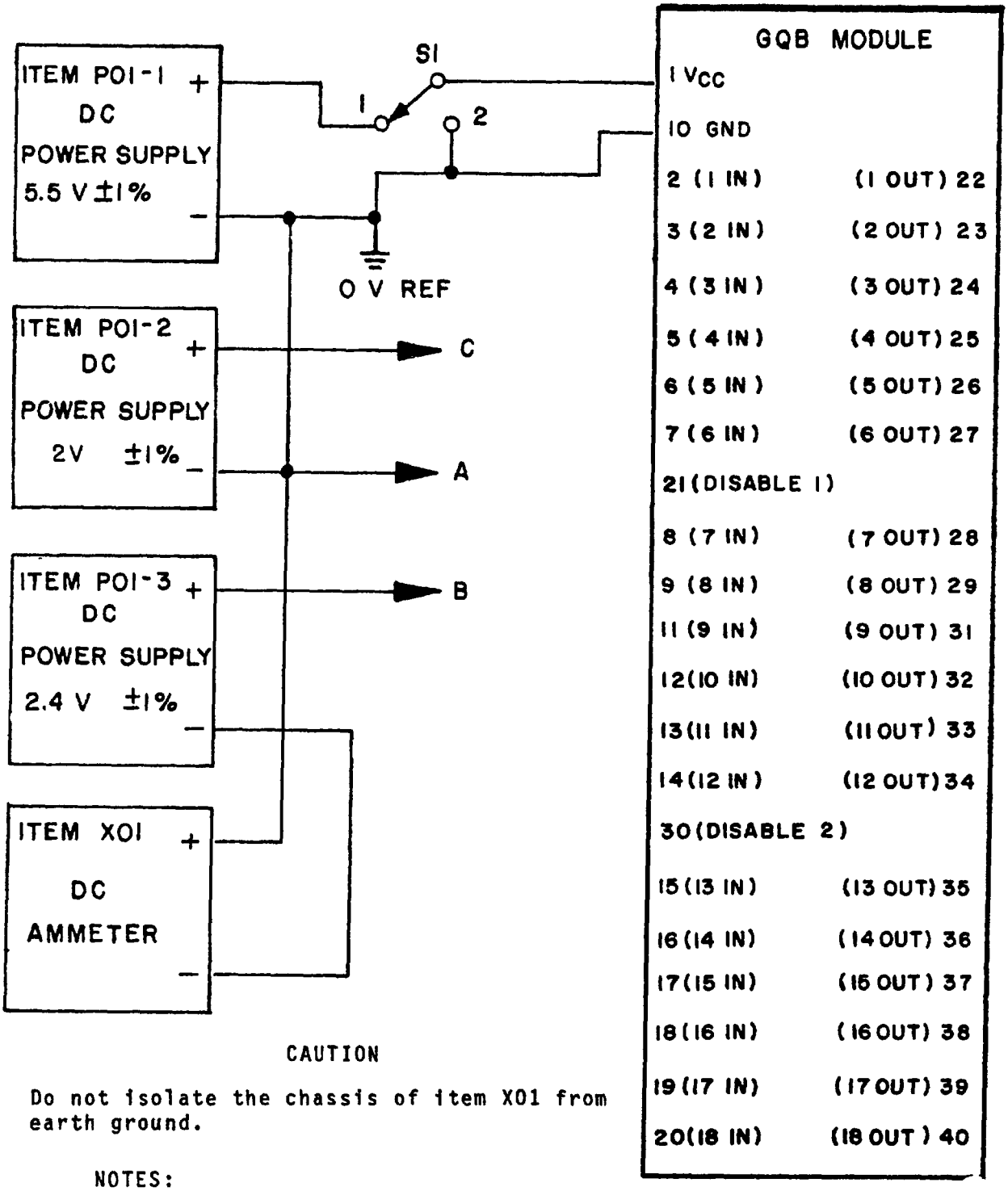
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
G1	1 OUT	Output leakage current, high level	2	22	21	-40	40	µA	10
G2	2 OUT	"	3	23	"	"	"	"	"
G3	3 OUT	"	4	24	"	"	"	"	"
G4	4 OUT	"	5	25	"	"	"	"	"
G5	5 OUT	"	6	26	"	"	"	"	"
G6	6 OUT	"	7	27	"	"	"	"	"
G7	7 OUT	"	8	28	30	"	"	"	"
G8	8 OUT	"	9	29	"	"	"	"	"
G9	9 OUT	"	11	31	"	"	"	"	"
G10	10 OUT	"	12	32	"	"	"	"	"
G11	11 OUT	"	13	33	"	"	"	"	"
G12	12 OUT	"	14	34	"	"	"	"	"
G13	13 OUT	"	15	35	"	"	"	"	"
G14	14 OUT	"	16	36	"	"	"	"	"
G15	15 OUT	"	17	37	"	"	"	"	"
G16	16 OUT	"	18	38	"	"	"	"	"
G17	17 OUT	"	19	39	"	"	"	"	"
G18	18 OUT	"	20	40	"	"	"	"	"

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TABLE IX. Output leakage current (power-off) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1 OUT	Output leakage current, power off	Open	22	Open	0	40	µA	10
H2	2 OUT	"	"	23	"	"	"	"	"
H3	3 OUT	"	"	24	"	"	"	"	"
H4	4 OUT	"	"	25	"	"	"	"	"
H5	5 OUT	"	"	26	"	"	"	"	"
H6	6 OUT	"	"	27	"	"	"	"	"
H7	7 OUT	"	"	28	"	"	"	"	"
H8	8 OUT	"	"	29	"	"	"	"	"
H9	9 OUT	"	"	31	"	"	"	"	"
H10	10 OUT	"	"	32	"	"	"	"	"
H11	11 OUT	"	"	33	"	"	"	"	"
H12	12 OUT	"	"	34	"	"	"	"	"
H13	13 OUT	"	"	35	"	"	"	"	"
H14	14 OUT	"	"	36	"	"	"	"	"
H15	15 OUT	"	"	37	"	"	"	"	"
H16	16 OUT	"	"	38	"	"	"	"	"
H17	17 OUT	"	"	39	"	"	"	"	"
H18	18 OUT	"	"	40	"	"	"	"	"

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. S1 = SPST switch.
2. For test condition G1 - G18: Switch 1 is in position 1.
3. For test condition H1 - H18: Switch 1 is in position 2.

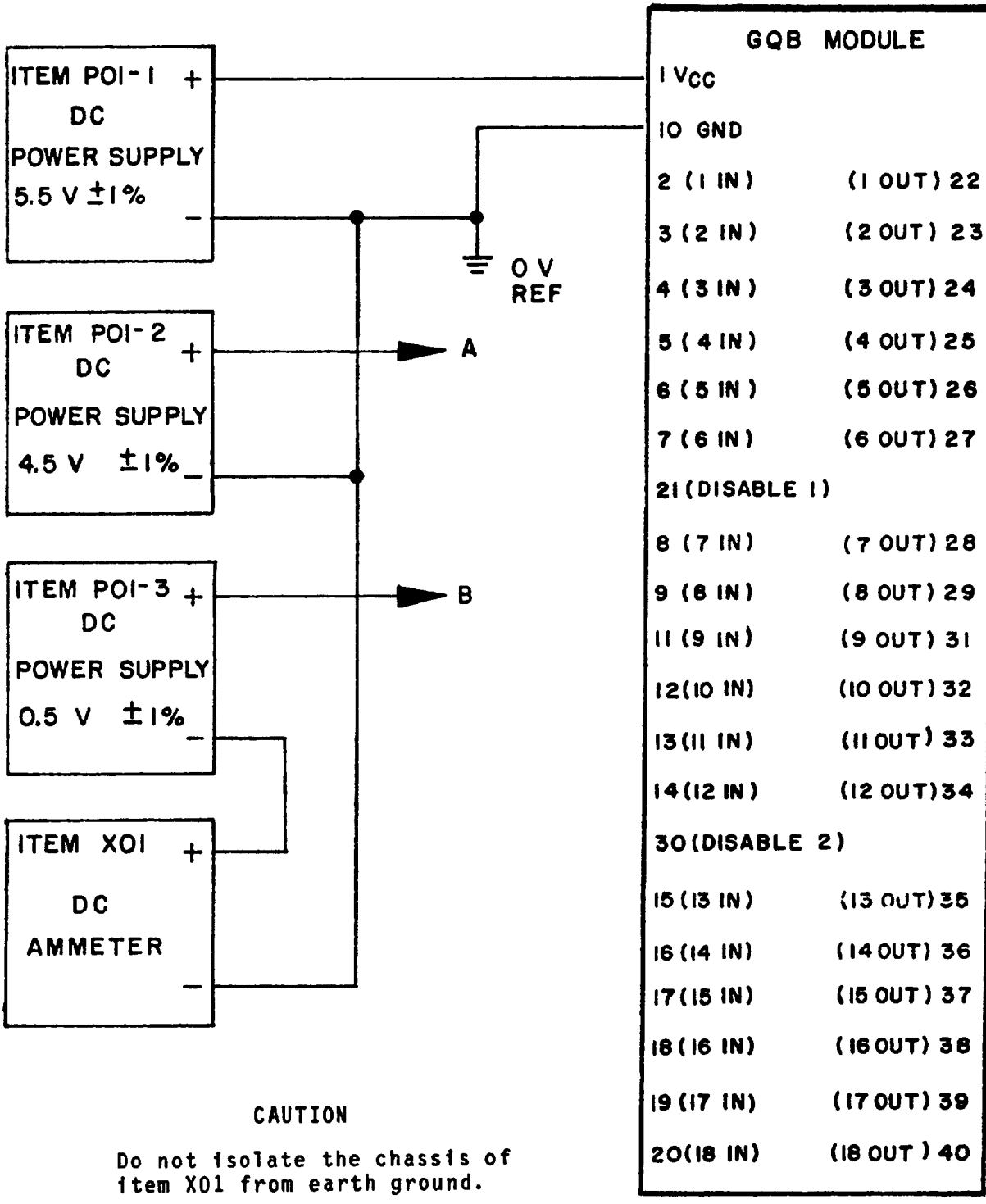
FIGURE 10. Output leakage current (high-level and power-off) test circuit for tables VIII and IX.

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TABLE X. Output leakage current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1 OUT	Output leakage current, low level	2, 21	22	-40	40	µA	11
J2	2 OUT	"	3, 21	23	"	"	"	"
J3	3 OUT	"	4, 21	24	"	"	"	"
J4	4 OUT	"	5, 21	25	"	"	"	"
J5	5 OUT	"	6, 21	26	"	"	"	"
J6	6 OUT	"	7, 21	27	"	"	"	"
J7	7 OUT	"	8, 30	28	"	"	"	"
J8	8 OUT	"	9, 30	29	"	"	"	"
J9	9 OUT	"	11, 30	31	"	"	"	"
J10	10 OUT	"	12, 30	32	"	"	"	"
J11	11 OUT	"	13, 30	33	"	"	"	"
J12	12 OUT	"	14, 30	34	"	"	"	"
J13	13 OUT	"	15, 30	35	"	"	"	"
J14	14 OUT	"	16, 30	36	"	"	"	"
J15	15 OUT	"	17, 30	37	"	"	"	"
J16	16 OUT	"	18, 30	38	"	"	"	"
J17	17 OUT	"	19, 30	39	"	"	"	"
J18	18 OUT	"	20, 30	40	"	"	"	"

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 11. Output leakage current (low-level) test circuit for table X.

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*TABLE XI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1 OUT Output voltage, low level (I_{OL} = 32 mA)	21, 2	Open	22	0	500	mV	12
K2	2 OUT "	21, 3	"	23	"	"	"	"
K3	3 OUT "	21, 4	"	24	"	"	"	"
K4	4 OUT "	21, 5	"	25	"	"	"	"
K5	5 OUT "	21, 6	"	26	"	"	"	"
K6	6 OUT "	21, 7	"	27	"	"	"	"
K7	7 OUT "	30, 8	"	28	"	"	"	"
K8	8 OUT "	30, 9	"	29	"	"	"	"
K9	9 OUT "	30, 11	"	31	"	"	"	"
K10	10 OUT "	30, 12	"	32	"	"	"	"
K11	11 OUT "	30, 13	"	33	"	"	"	"
K12	12 OUT "	30, 14	"	34	"	"	"	"
K13	13 OUT "	30, 15	"	35	"	"	"	"
K14	14 OUT "	30, 16	"	36	"	"	"	"
K15	15 OUT "	30, 17	"	37	"	"	"	"
K16	16 OUT "	30, 18	"	38	"	"	"	"
K17	17 OUT "	30, 19	"	39	"	"	"	"
K18	18 OUT "	30, 20	"	40	"	"	"	"

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TABLE XII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1 OUT Output voltage, high level	3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,30	2	22	2.4	4.5	V	12
L2	2 OUT "	2,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,30	3	23	"	"	"	"
L3	3 OUT "	2,3,5,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,30	4	24	"	"	"	"
L4	4 OUT "	2,3,4,6,7,8,9,11,12,13,14,15,16,17,18,19,20,21,30	5	25	"	"	"	"
L5	5 OUT "	2,3,4,5,7,8,9,11,12,13,14,15,16,17,18,19,20,21,30	6	26	"	"	"	"
L6	6 OUT "	2,3,4,5,6,8,9,11,12,13,14,15,16,17,18,19,20,21,30	7	27	"	"	"	"

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TABLE XII. Output voltage (high-level) test - Continued.

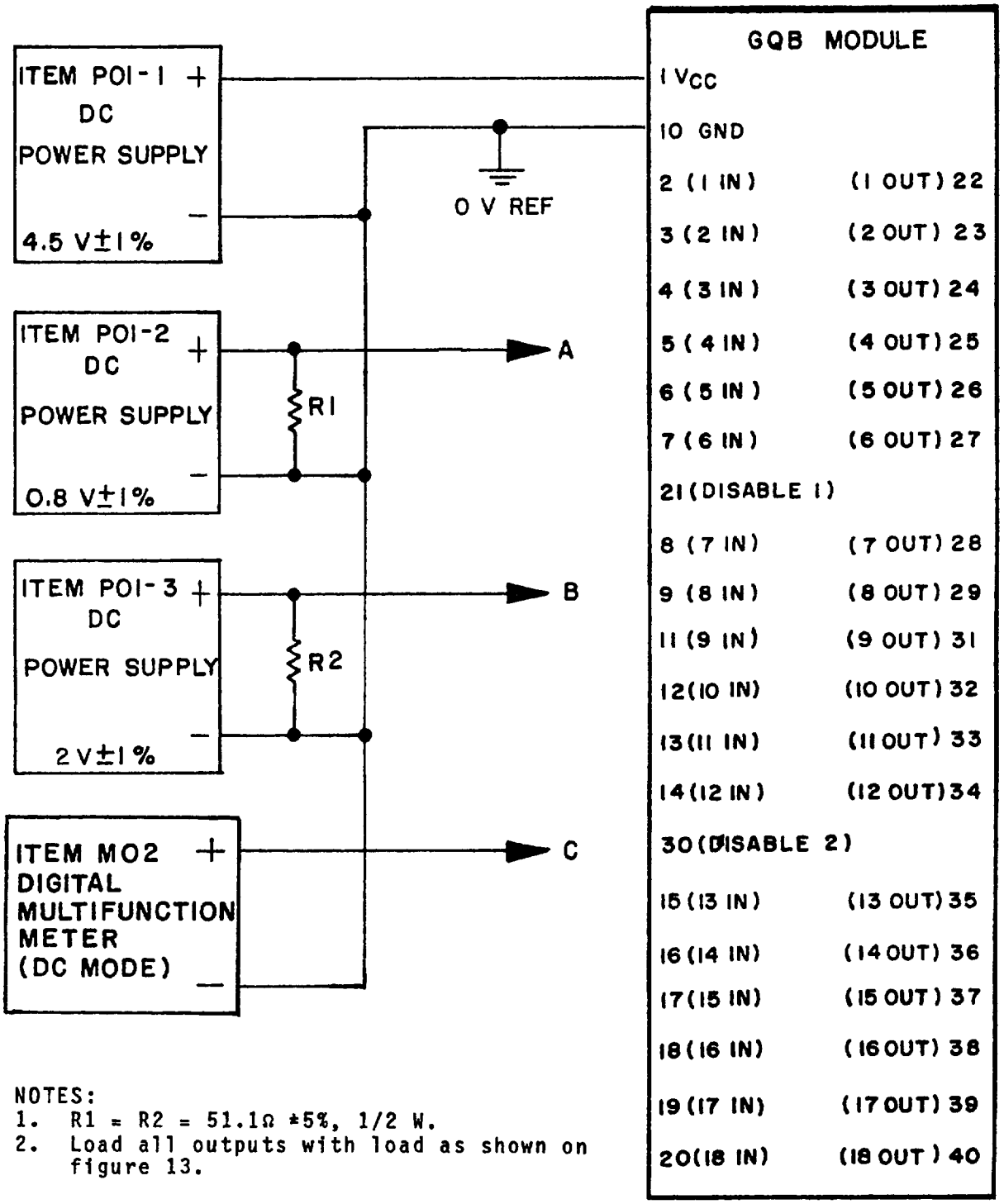
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L7	7 OUT Output voltage, high level	2,3,4,5,6,7,9,11,12,13,14,15,16,17,18,19,20,21,30	8	28	2.4	4.5	V	12
L8	8 OUT "	2,3,4,5,6,7,8,11,12,13,14,15,16,17,18,19,20,21,30	9	29	"	"	"	"
L9	9 OUT "	2,3,4,5,6,7,8,9,12,13,14,15,16,17,18,19,20,21,30	11	31	"	"	"	"
L10	10 OUT "	2,3,4,5,6,7,8,9,11,13,14,15,16,17,18,19,20,21,30	12	32	"	"	"	"
L11	11 OUT "	2,3,4,5,6,7,8,9,11,12,14,15,16,17,18,19,20,21,30	13	33	"	"	"	"
L12	12 OUT "	2,3,4,5,6,7,8,9,11,12,13,15,16,17,18,19,20,21,30	14	34	"	"	"	"

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TABLE XII. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L13	13 OUT	Output voltage, high level	2,3,4,5,6,7,8,9,11,12,13,14,16,17,18,19,20,21,30	15	35	2.4	4.5	V	12
L14	14 OUT	"	2,3,4,5,6,7,8,9,11,12,13,14,15,17,18,19,20,21,30	16	36	"	"	"	"
L15	15 OUT	"	2,3,4,5,6,7,8,9,11,12,13,14,15,16,18,19,20,21,30	17	37	"	"	"	"
L16	16 OUT	"	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,19,20,21,30	18	38	"	"	"	"
L17	17 OUT	"	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,20,21,30	19	39	"	"	"	"
L18	18 OUT	"	2,3,4,5,6,7,8,9,11,12,13,14,15,16,17,18,19,21,30	20	40	"	"	"	"

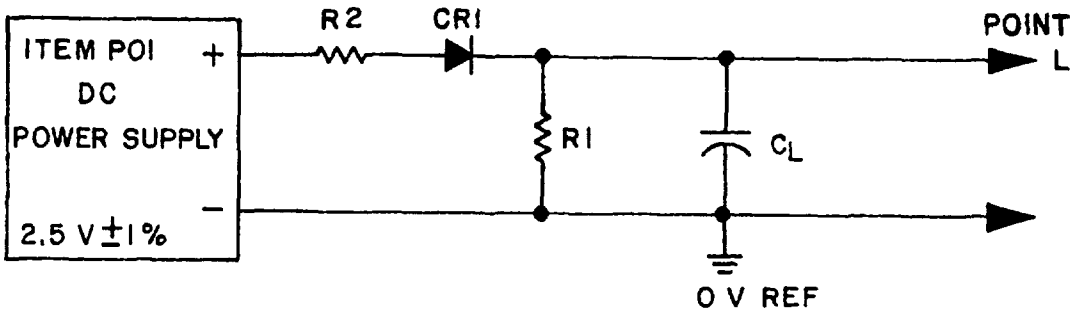
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Load all outputs with load as shown on figure 13.

* FIGURE 12. Output voltage (low-level and high-level) test circuit for tables XI and XII.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 50 pF ±5% (including probe and parasitics) for test conditions Q1-Q18, R1-R18, S1-S18, T1-T18, V1-V18, X1-X18, Y1-Y8, and Z1-Z18.
3. R1 = 120Ω ±1%, 1/8 W (current flow of 2 mA when point L is held at 2.4 V).
4. R2 = Select value for a current flow of 32 mA ±5% out of point L with point L held at 0.5 V.
5. During output switching, the 2.5 V supply shall have a maximum overshoot of 250 mV.
6. C_L = 1,000 pF ±5% (including probe and parasitic capacitance) for test conditions S19-S36 and T19-T36.
7. Capacitor C_L is not required for test conditions K and L.
8. Lead length from load circuit to module output pin under test shall be 1.00 inch (25.4 mm) or less.
9. $C_L \leq$ 10 pF for test conditions U1-U18 and W1-W18.

FIGURE 13. GQB logic load circuit.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1 IN Input current, high level	Open	2	3,4,5,6,7,8, 9,11,12,13, 14,15,16,17, 18,19,20	Open	0	40	µA	14
M2	2 IN "	"	3	2,4,5,6,7,8, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M3	3 IN "	"	4	2,3,5,6,7,8, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M4	4 IN "	"	5	2,3,4,6,7,8, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M5	5 IN "	"	6	2,3,4,5,7,8, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M6	6 IN "	"	7	2,3,4,5,6,8, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M7	7 IN "	"	8	2,3,4,5,6,7, 9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M8	8 IN Input current, high level	Open	9	2,3,4,5,6,7, 8,11,12,13, 14,15,16,17, 18,19,20	Open	0	40	µA	14
M9	9 IN "	"	11	2,3,4,5,6,7, 8,9,12,13,14, 15,16,17,18, 19,20	"	"	"	"	"
M10	10 IN "	"	12	2,3,4,5,6,7, 8,9,11,13,14, 15,16,17,18, 19,20	"	"	"	"	"
M11	11 IN "	"	13	2,3,4,5,6,7, 8,9,11,12,14, 15,16,17,18, 19,20	"	"	"	"	"
M12	12 IN "	"	14	2,3,4,5,6,7, 8,9,11,12,13, 15,16,17,18, 19,20	"	"	"	"	"
M13	13 IN "	"	15	2,3,4,5,6,7, 8,9,11,12,13, 14,16,17,18, 19,20	"	"	"	"	"
M14	14 IN "	"	16	2,3,4,5,6,7, 8,9,11,12,13, 14,15,17,18, 19,20	"	"	"	"	"

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M15	15 IN Input current, high level	Open	17	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,18, 19,20	Open	0	40	µA	14
M16	16 IN "	"	18	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 19,20	"	"	"	"	"
M17	17 IN "	"	19	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,20	"	"	"	"	"
M18	18 IN "	"	20	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,19	"	"	"	"	"
M19	Disable 1 "	"	21	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,19,20	"	"	"	"	"
M20	Disable 2 "	"	30	2,3,4,5,6,7, 8,9,11,12,13, 14,15,16,17, 18,19,20	"	"	80	"	"

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*TABLE XIV. Input current (low-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1 IN	Input current, low level (disable low)	2	Open	21	Open	0	-1.76	mA	14
N2	2 IN	"	3	"	"	"	"	"	"	"
N3	3 IN	"	4	"	"	"	"	"	"	"
N4	4 IN	"	5	"	"	"	"	"	"	"
N5	5 IN	"	6	"	"	"	"	"	"	"
N6	6 IN	"	7	"	"	"	"	"	"	"
N7	7 IN	"	8	"	30	"	"	"	"	"
N8	8 IN	"	9	"	"	"	"	"	"	"
N9	9 IN	"	11	"	"	"	"	"	"	"
N10	10 IN	"	12	"	"	"	"	"	"	"
N11	11 IN	"	13	"	"	"	"	"	"	"
N12	12 IN	"	14	"	"	"	"	"	"	"
N13	13 IN	"	15	"	"	"	"	"	"	"
N14	14 IN	"	16	"	"	"	"	"	"	"
N15	15 IN	"	17	"	"	"	"	"	"	"
N16	16 IN	"	18	"	"	"	"	"	"	"
N17	17 IN	"	19	"	"	"	"	"	"	"

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*TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N18	18 IN	Input current, low level (disable low)	20	Open	30	Open	0	-1.76	mA	14
N19	DISABLE 1	"	21	"	Open	"	"	-1.76	"	"
N20	DISABLE 2	"	30	"	"	"	"	-3.52	"	"
N21	1 IN	Input current, low level (disable high)	2	"	"	21	"	-1.76	"	"
N22	2 IN	"	3	"	"	"	"	"	"	"
N23	3 IN	"	4	"	"	"	"	"	"	"
N24	4 IN	"	5	"	"	"	"	"	"	"
N25	5 IN	"	6	"	"	"	"	"	"	"
N26	6 IN	"	7	"	"	"	"	"	"	"
N27	7 IN	"	8	"	"	30	"	"	"	"
N28	8 IN	"	9	"	"	"	"	"	"	"
N29	9 IN	"	11	"	"	"	"	"	"	"
N30	10 IN	"	12	"	"	"	"	"	"	"
N31	11 IN	"	13	"	"	"	"	"	"	"
N32	12 IN	"	14	"	"	"	"	"	"	"

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*TABLE XIV. Input current (low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N33	13 IN	Input current, low level (disable high)	15	Open	Open	30	0	-1.76	mA	14
N34	14 IN	"	16	"	"	"	"	"	"	"
N35	15 IN	"	17	"	"	"	"	"	"	"
N36	16 IN	"	18	"	"	"	"	"	"	"
N37	17 IN	"	19	"	"	"	"	"	"	"
N38	18 IN	"	20	"	"	"	"	"	"	"

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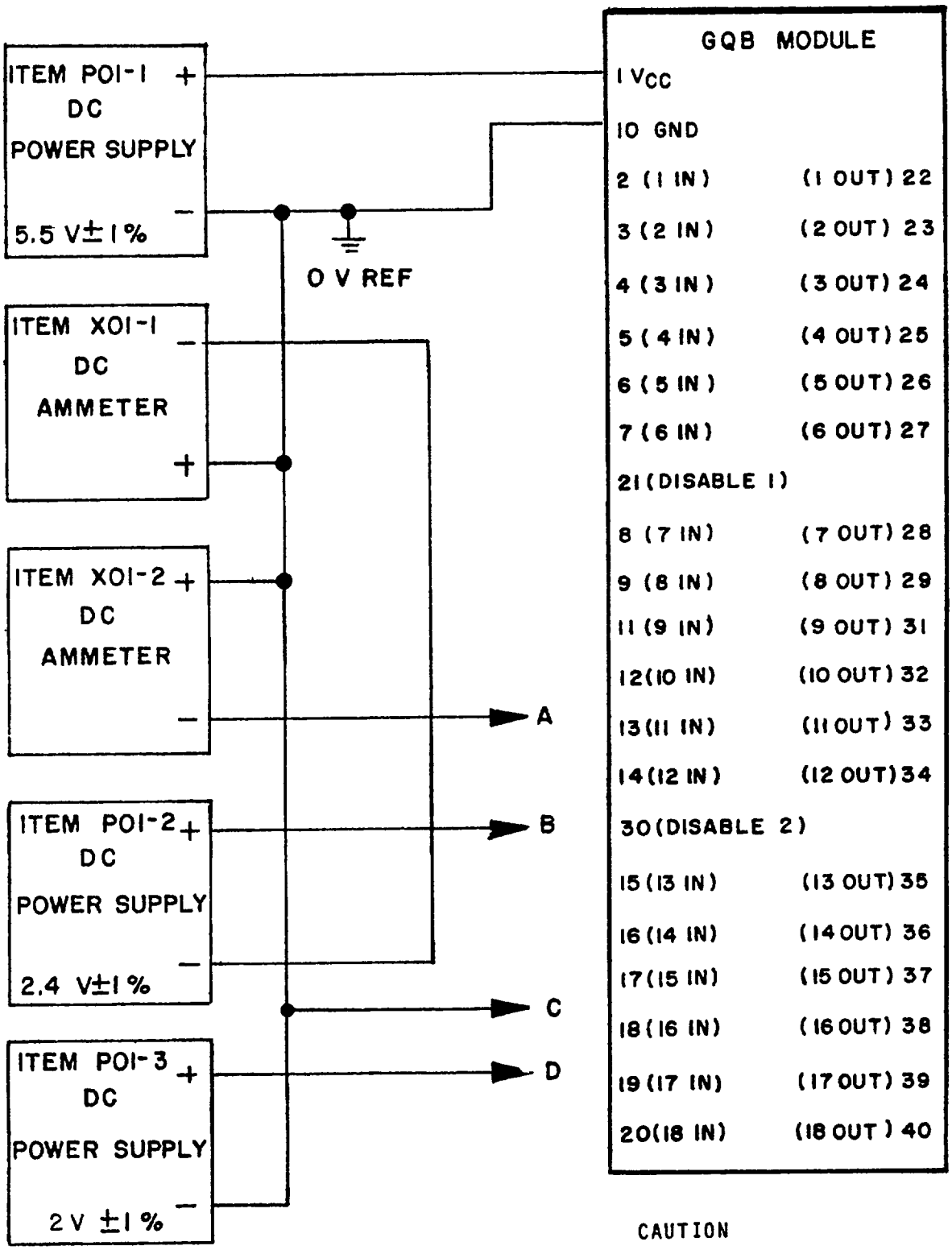


FIGURE 14. Input current (low-level and high-level) test circuit for tables XIII and XIV.

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NOTES:
1. Since items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required.
2. With point B disconnected, item XO1-1 should indicate 0.0 current on the 1 μA scale.

FIGURE 14. Input current (low-level and high-level) test circuit for tables XIII and XIV - Continued.

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TABLE XV. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply current	60	294	mA	15

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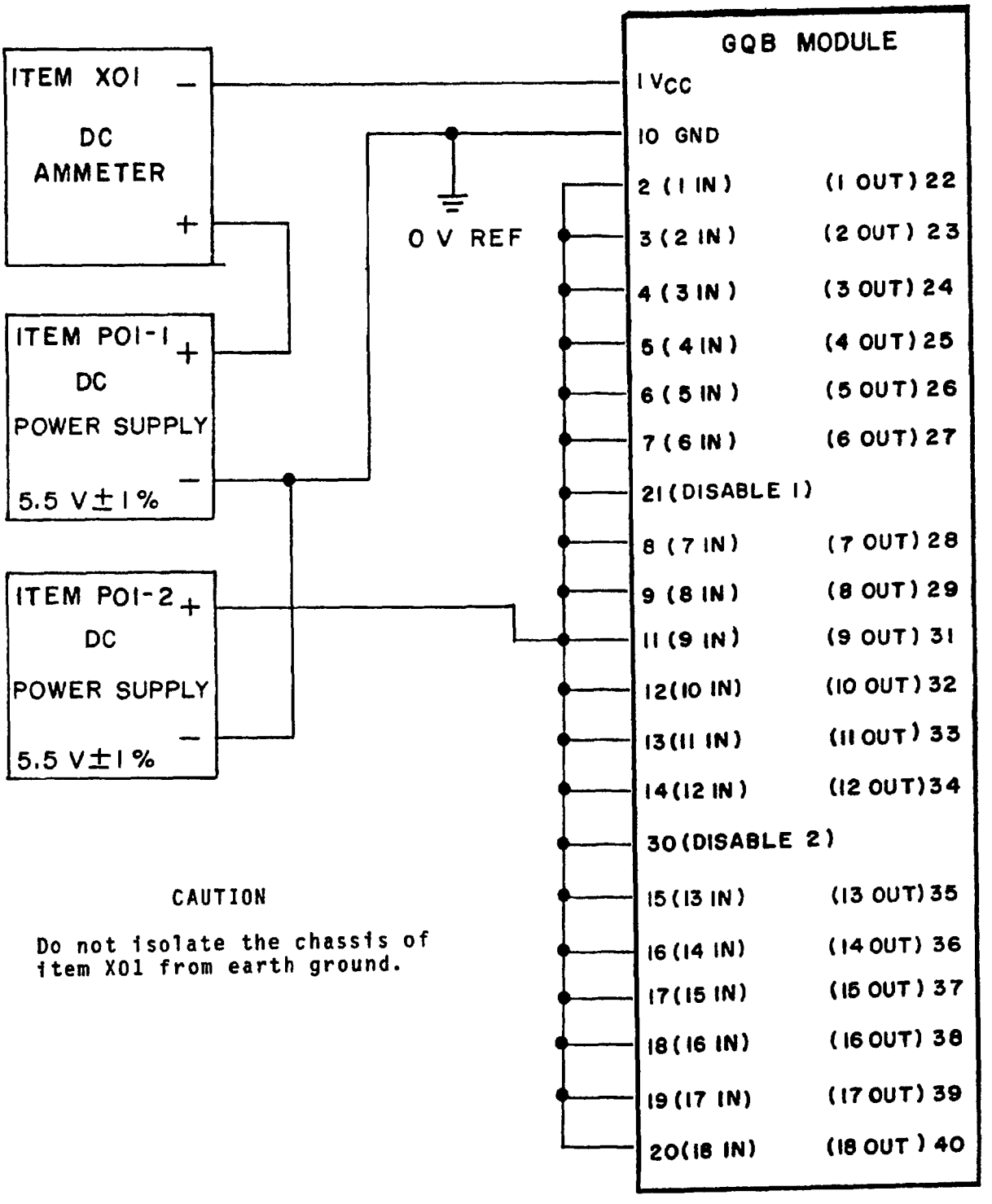


FIGURE 15. Power supply current test circuit for table XV.

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*TABLE XVI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect			Test cond Q Initial limits 25°C ±5°C ambient		Test cond R Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
Q1	1 OUT Transition time (t_{TLH}), C_L = 50 pF	2	22	21	4	40	---	---	ns	16, 17
R1	1 OUT Transition time (t_{THL}), C_L = 50 pF	2	22	"	---	---	2	9	"	"
Q2	2 OUT Transition time (t_{TLH}), C_L = 50 pF	3	23	"	4	40	---	---	"	"
R2	2 OUT Transition time (t_{THL}), C_L = 50 pF	3	23	"	---	---	2	9	"	"
Q3	3 OUT Transition time (t_{TLH}), C_L = 50 pF	4	24	"	4	40	---	---	"	"
R3	3 OUT Transition time (t_{THL}), C_L = 50 pF	4	24	"	---	---	2	9	"	"
Q4	4 OUT Transition time (t_{TLH}), C_L = 50 pF	5	25	"	4	40	---	---	"	"
R4	4 OUT Transition time (t_{THL}), C_L = 50 pF	5	25	"	---	---	2	9	"	"
Q5	5 OUT Transition time (t_{TLH}), C_L = 50 pF	6	26	"	4	40	---	---	"	"
R5	5 OUT Transition time (t_{THL}), C_L = 50 pF	6	26	"	---	---	2	9	"	"
Q6	6 OUT Transition time (t_{TLH}), C_L = 50 pF	7	27	"	4	40	---	---	"	"
R6	6 OUT Transition time (t_{THL}), C_L = 50 pF	7	27	"	---	---	2	9	"	"

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*TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Test cond Q Initial limits 25°C ±5°C ambient		Test cond R Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
Q7	7 OUT Transition time (t_{TLH}), C_L = 50 pF	8	28	30	4	40	---	---	ns	16, 17
R7	7 OUT Transition time (t_{THL}), C_L = 50 pF	8	28	"	---	---	2	9	"	"
Q8	8 OUT Transition time (t_{TLH}), C_L = 50 pF	9	29	"	4	40	---	---	"	"
R8	8 OUT Transition time (t_{THL}), C_L = 50 pF	9	29	"	---	---	2	9	"	"
Q9	9 OUT Transition time (t_{TLH}), C_L = 50 pF	11	31	"	4	40	---	---	"	"
R9	9 OUT Transition time (t_{THL}), C_L = 50 pF	11	31	"	---	---	2	9	"	"
Q10	10 OUT Transition time (t_{TLH}), C_L = 50 pF	12	32	"	4	40	---	---	"	"
R10	10 OUT Transition time (t_{THL}), C_L = 50 pF	12	32	"	---	---	2	9	"	"
Q11	11 OUT Transition time (t_{TLH}), C_L = 50 pF	13	33	"	4	40	---	---	"	"
R11	11 OUT Transition time (t_{THL}), C_L = 50 pF	13	33	"	---	---	2	9	"	"
Q12	12 OUT Transition time (t_{TLH}), C_L = 50 pF	14	34	"	4	40	---	---	"	"
R12	12 OUT Transition time (t_{THL}), C_L = 50 pF	14	34	"	---	---	2	9	"	"

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*TABLE XVI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect			Test cond Q Initial limits 25°C ±5°C ambient		Test cond R Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
Q13	13 OUT Transition time (t_{TLH}), C_L = 50 pF	15	35	30	4	40	---	---	ns	16, 17
R13	13 OUT Transition time (t_{THL}), C_L = 50 pF	15	35	"	---	---	2	9	"	"
Q14	14 OUT Transition time (t_{TLH}), C_L = 50 pF	16	36	"	4	40	---	---	"	"
R14	14 OUT Transition time (t_{THL}), C_L = 50 pF	16	36	"	---	---	2	9	"	"
Q15	15 OUT Transition time (t_{TLH}), C_L = 50 pF	17	37	"	4	40	---	---	"	"
R15	15 OUT Transition time (t_{THL}), C_L = 50 pF	17	37	"	---	---	2	9	"	"
Q16	16 OUT Transition time (t_{TLH}), C_L = 50 pF	18	38	"	4	40	---	---	"	"
R16	16 OUT Transition time (t_{THL}), C_L = 50 pF	18	38	"	---	---	2	9	"	"
Q17	17 OUT Transition time (t_{TLH}), C_L = 50 pF	19	39	"	4	40	---	---	"	"
R17	17 OUT Transition time (t_{THL}), C_L = 50 pF	19	39	"	---	---	2	9	"	"
Q18	18 OUT Transition time (t_{TLH}), C_L = 50 pF	20	40	"	4	40	---	---	"	"
R18	18 OUT Transition time (t_{THL}), C_L = 50 pF	20	40	"	---	---	2	9	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S1	1 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	2	22	21	3	16	---	---	ns	16, 17
T1	1 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	2	22	"	---	---	3	22	"	"
S2	2 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	3	23	"	3	16	---	---	"	"
T2	2 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	3	23	"	---	---	3	22	"	"
S3	3 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	4	24	"	3	16	---	---	"	"
T3	3 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	4	24	"	---	---	3	22	"	"
S4	4 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	5	25	"	3	16	---	---	"	"
T4	4 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	5	25	"	---	---	3	22	"	"
S5	5 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	6	26	"	3	16	---	---	"	"
T5	5 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	6	26	"	---	---	3	22	"	"
S6	6 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	7	27	"	3	16	---	---	"	"
T6	6 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	7	27	"	---	---	3	22	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S7	7 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	8	28	30	3	16	---	---	ns	16, 17
T7	7 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	8	28	"	---	---	3	22	"	"
S8	8 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	9	29	"	3	16	---	---	"	"
T8	8 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	9	29	"	---	---	3	22	"	"
S9	9 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	11	31	"	3	16	---	---	"	"
T9	9 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	11	31	"	---	---	3	22	"	"
S10	10 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	12	32	"	3	16	---	---	"	"
T10	10 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	12	32	"	---	---	3	22	"	"
S11	11 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	13	33	"	3	16	---	---	"	"
T11	11 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	13	33	"	---	---	3	22	"	"
S12	12 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	14	34	"	3	16	---	---	"	"
T12	12 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	14	34	"	---	---	3	22	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S13	13 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	15	35	30	3	16	---	---	ns	16, 17
T13	13 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	15	35	"	---	---	3	22	"	"
S14	14 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	16	36	"	3	16	---	---	"	"
T14	14 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	16	36	"	---	---	3	22	"	"
S15	15 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	17	37	"	3	16	---	---	"	"
T15	15 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	17	37	"	---	---	3	22	"	"
S16	16 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	18	38	"	3	16	---	---	"	"
T16	16 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	18	38	"	---	---	3	22	"	"
S17	17 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	19	39	"	3	16	---	---	"	"
T17	17 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	19	39	"	---	---	3	22	"	"
S18	18 OUT Propagation delay time (t_{PLH}), C_L = 50 pF	20	40	"	3	16	---	---	"	"
T18	18 OUT Propagation delay time (t_{PHL}), C_L = 50 pF	20	40	"	---	---	3	22	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S19	1 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	2	22	21	8	63	---	---	ns	16, 17
T19	1 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	2	22	"	---	---	8	70	"	"
S20	2 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	3	23	"	8	63	---	---	"	"
T20	2 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	3	23	"	---	---	8	70	"	"
S21	3 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	4	24	"	8	63	---	---	"	"
T21	3 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	4	24	"	---	---	8	70	"	"
S22	4 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	5	25	"	8	63	---	---	"	"
T22	4 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	5	25	"	---	---	8	70	"	"
S23	5 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	6	26	"	8	63	---	---	"	"
T23	5 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	6	26	"	---	---	8	70	"	"
S24	6 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	7	27	"	8	63	---	---	"	"
T24	6 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	7	27	"	---	---	8	70	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S25	7 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	8	28	30	8	63	---	---	ns	16, 17
T25	7 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	8	28	"	---	---	8	70	"	"
S26	8 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	9	29	"	8	63	---	---	"	"
T26	8 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	9	29	"	---	---	8	70	"	"
S27	9 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	11	31	"	8	63	---	---	"	"
T27	9 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	11	31	"	---	---	8	70	"	"
S28	10 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	12	32	"	8	63	---	---	"	"
T28	10 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	12	32	"	---	---	8	70	"	"
S29	11 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	13	33	"	8	63	---	---	"	"
T29	11 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	13	33	"	---	---	8	70	"	"
S30	12 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	14	34	"	8	63	---	---	"	"
T30	12 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	14	34	"	---	---	8	70	"	"

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TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Test cond S Initial limits 25°C ±5°C ambient		Test cond T Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
S31	13 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	15	35	30	8	63	---	---	ns	16, 17
T31	13 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	15	35	"	---	---	8	70	"	"
S32	14 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	16	36	"	8	63	---	---	"	"
T32	14 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	16	36	"	---	---	8	70	"	"
S33	15 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	17	37	"	8	63	---	---	"	"
T33	15 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	17	37	"	---	---	8	70	"	"
S34	16 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	18	38	"	8	63	---	---	"	"
T34	16 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	18	38	"	---	---	8	70	"	"
S35	17 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	19	39	"	8	63	---	---	"	"
T35	17 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	19	39	"	---	---	8	70	"	"
S36	18 OUT Propagation delay time (t_{PLH}), C_L = 1000 pF	20	40	"	8	63	---	---	"	"
T36	18 OUT Propagation delay time (t_{PHL}), C_L = 1000 pF	20	40	"	---	---	8	70	"	"

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TABLE XVIII. Disable delay time (t_{oh} and t_{ho}) test.

Test condition	Parameter	Connect			Test cond U Initial limits 25°C ±5°C ambient		Test cond V Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
U1	1 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	21	22	2	2	27	---	---	ns	16, 17
V1	1 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	22	2	---	---	2	37	"	"
U2	2 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	23	3	2	27	---	---	"	"
V2	2 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	23	3	---	---	2	37	"	"
U3	3 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	24	4	2	27	---	---	"	"
V3	3 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	24	4	---	---	2	37	"	"
U4	4 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	25	5	2	27	---	---	"	"
V4	4 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	25	5	---	---	2	37	"	"
U5	5 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	26	6	2	27	---	---	"	"
V5	5 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	26	6	---	---	2	37	"	"
U6	6 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	27	7	2	27	---	---	"	"
V6	6 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	27	7	---	---	2	37	"	"

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TABLE XVIII. Disable delay time (t_{oh} and t_{ho}) test - Continued.

Test condition	Parameter	Connect			Test cond U Initial limits 25°C ±5°C ambient		Test cond V Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max	Min	Max		
U7	7 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	30	28	8	2	27	---	---	ns	16, 17
V7	7 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	28	8	---	---	2	37	"	"
U8	8 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	29	9	2	27	---	---	"	"
V8	8 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	29	9	---	---	2	37	"	"
U9	9 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	31	11	2	27	---	---	"	"
V9	9 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	31	11	---	---	2	37	"	"
U10	10 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	32	12	2	27	---	---	"	"
V10	10 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	32	12	---	---	2	37	"	"
U11	11 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	33	13	2	27	---	---	"	"
V11	11 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	33	13	---	---	2	37	"	"
U12	12 OUT Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	34	14	2	27	---	---	"	"
V12	12 OUT Disable delay time (t_{ho}), $C_L = 50$ pF	"	34	14	---	---	2	37	"	"

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TABLE XVIII. Disable delay time (t_{oh} and t_{ho}) test - Continued.

Test condition	Parameter		Connect			Test cond U Initial limits 25°C ±5°C ambient		Test cond V Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max	Min	Max		
U13	13 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	30	35	15	2	27	---	---	ns	16, 17
V13	13 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	35	15	---	---	2	37	"	"
U14	14 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	36	16	2	27	---	---	"	"
V14	14 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	36	16	---	---	2	37	"	"
U15	15 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	37	17	2	27	---	---	"	"
V15	15 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	37	17	---	---	2	37	"	"
U16	16 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	38	18	2	27	---	---	"	"
V16	16 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	38	18	---	---	2	37	"	"
U17	17 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	39	19	2	27	---	---	"	"
V17	17 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	39	19	---	---	2	37	"	"
U18	18 OUT	Disable delay time (t_{oh}), $C_L \leq 10$ pF	"	40	20	2	27	---	---	"	"
V18	18 OUT	Disable delay time (t_{ho}), $C_L = 50$ pF	"	40	20	---	---	2	37	"	"

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*TABLE XIX. Disable delay time (t_{1h} and t_{h1}) test.

Test condition	Parameter	Connect				Test cond W Initial limits 25°C ±5°C ambient		Test cond X Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
W1	1 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	21	22	Open	2	2	16	---	---	ns	16, 17
X1	1 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	22	"	2	---	---	2	40	"	"
W2	2 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	23	"	3	2	16	---	---	"	"
X2	2 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	23	"	3	---	---	2	40	"	"
W3	3 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	24	"	4	2	16	---	---	"	"
X3	3 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	24	"	4	---	---	2	40	"	"
W4	4 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	25	"	5	2	16	---	---	"	"
X4	4 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	25	"	5	---	---	2	40	"	"
W5	5 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	26	"	6	2	16	---	---	"	"
X5	5 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	26	"	6	---	---	2	40	"	"
W6	6 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	27	"	7	2	16	---	---	"	"
X6	6 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	27	"	7	---	---	2	40	"	"

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*TABLE XIX. Disable delay time (t_{1h} and t_{h1}) test - Continued.

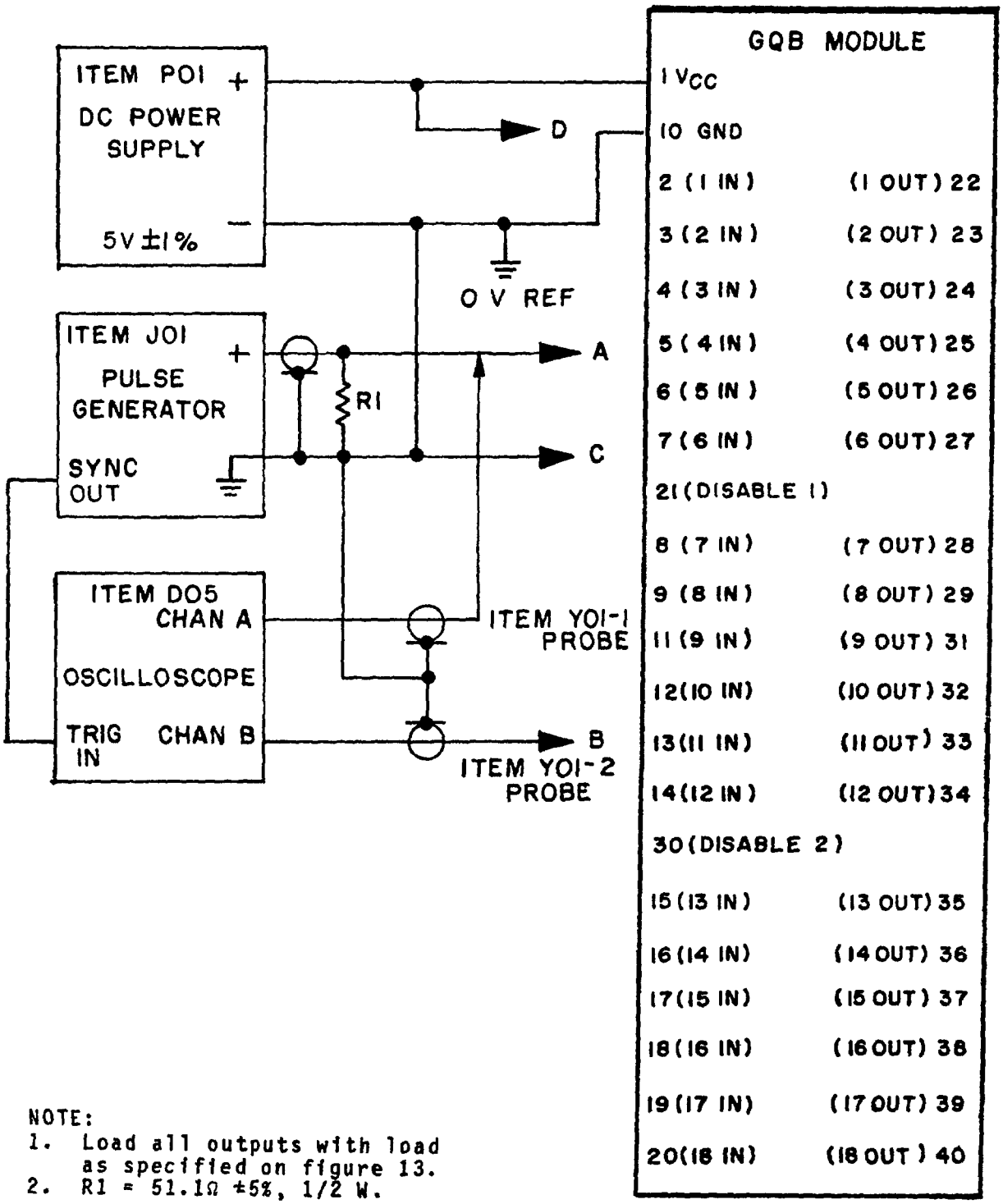
Test condition	Parameter	Connect				Test cond W Initial limits 25°C ±5°C ambient		Test cond X Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
W7	7 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	30	28	Open	8	2	16	---	---	ns	16, 17
X7	7 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	28	"	8	---	---	2	40	"	"
W8	8 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	29	"	9	2	16	---	---	"	"
X8	8 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	29	"	9	---	---	2	40	"	"
W9	9 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	31	"	11	2	16	---	---	"	"
X9	9 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	31	"	11	---	---	2	40	"	"
W10	10 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	32	"	12	2	16	---	---	"	"
X10	10 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	32	"	12	---	---	2	40	"	"
W11	11 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	33	"	13	2	16	---	---	"	"
X11	11 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	33	"	13	---	---	2	40	"	"
W12	12 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	34	"	14	2	16	---	---	"	"
X12	12 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	34	"	14	---	---	2	40	"	"

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*TABLE XIX. Disable delay time (t_{1h} and t_{h1}) test - Continued.

Test condition	Parameter	Connect				Test cond W Initial limits 25°C ±5°C ambient		Test cond X Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max	Min	Max		
W13	13 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	30	35	Open	15	2	16	---	---	ns	16, 17
X13	13 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	35	"	15	---	---	2	40	"	"
W14	14 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	36	"	16	2	16	---	---	"	"
X14	14 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	36	"	16	---	---	2	40	"	"
W15	15 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	37	"	17	2	16	---	---	"	"
X15	15 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	37	"	17	---	---	2	40	"	"
W16	16 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	38	"	18	2	16	---	---	"	"
X16	16 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	38	"	18	---	---	2	40	"	"
W17	17 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	39	"	19	2	16	---	---	"	"
X17	17 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	39	"	19	---	---	2	40	"	"
W18	18 OUT Disable delay time (t_{1h}), $C_L \leq 10$ pF	"	40	"	20	2	16	---	---	"	"
X18	18 OUT Disable delay time (t_{h1}), $C_L = 50$ pF	"	40	"	20	---	---	2	40	"	"

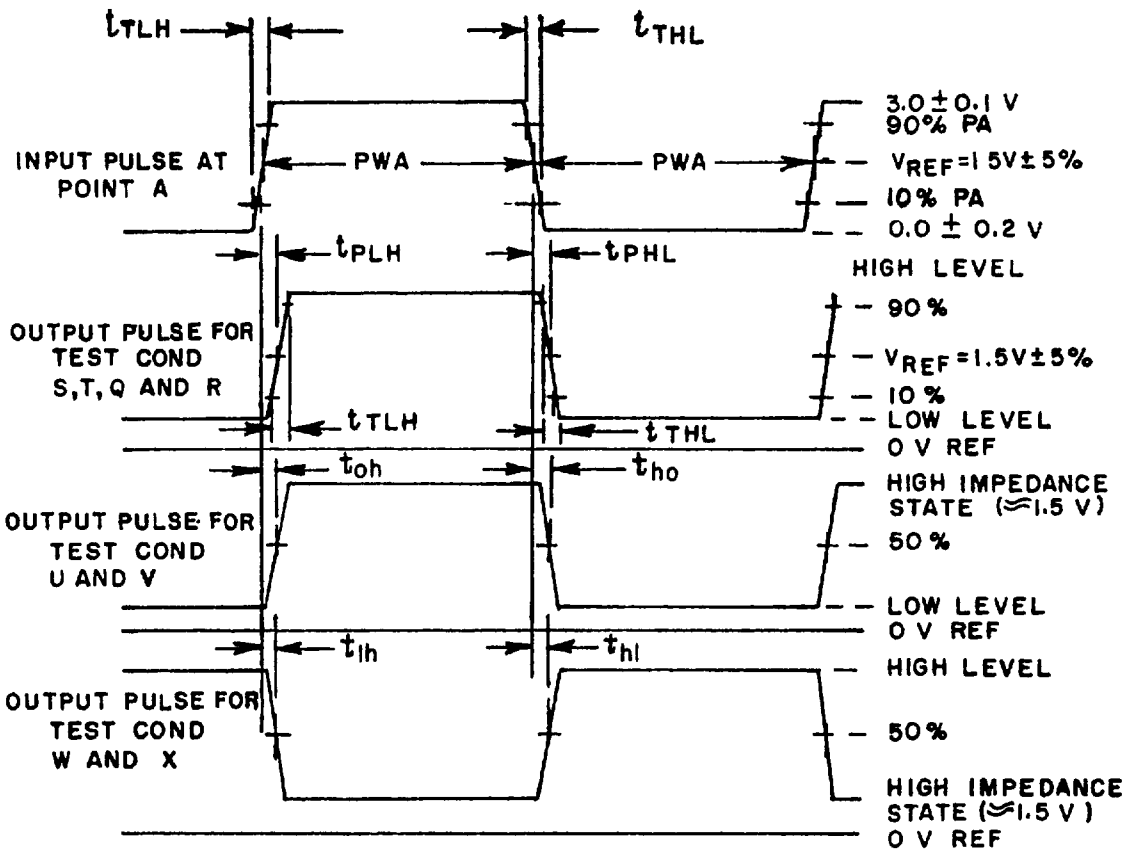
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NOTE:
1. Load all outputs with load as specified on figure 13.
2. R1 = 51.1Ω ±5%, 1/2 W.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and disable delay time (t_{0h}, t_{h0}, t_{1h}, and t_{h1}) test circuit for tables XVI, XVII, XVIII, and XIX.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making measurements, verify that the output high level is ≥ 2.4 V and the low level is ≤ 0.5 V unless otherwise specified.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and disable delay time (t_{oh}, t_{ho}, t_{lh}, and t_{hl}) test circuit waveforms for tables XVI, XVII, XVIII, and XIX.

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*TABLE XX. Output crosstalk (high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	1 OUT	Output crosstalk, high level	Open	2	Group A 1/	22	0	700	mV	18,19
Y2	2 OUT	"	"	3	"	23	"	"	"	"
Y3	3 OUT	"	"	4	"	24	"	"	"	"
Y4	4 OUT	"	"	5	"	25	"	"	"	"
Y5	5 OUT	"	"	6	"	26	"	"	"	"
Y6	6 OUT	"	"	7	"	27	"	"	"	"
Y7	7 OUT	"	"	8	"	28	"	"	"	"
Y8	8 OUT	"	"	9	"	29	"	"	"	"
Y9	9 OUT	"	"	11	"	31	"	"	"	"
Y10	10 OUT	"	"	12	"	32	"	"	"	"
Y11	11 OUT	"	"	13	"	33	"	"	"	"
Y12	12 OUT	"	"	14	"	34	"	"	"	"
Y13	13 OUT	"	"	15	"	35	"	"	"	"
Y14	14 OUT	"	"	16	"	36	"	"	"	"
Y15	15 OUT	"	"	17	"	37	"	"	"	"
Y16	16 OUT	"	"	18	"	38	"	"	"	"
Y17	17 OUT	"	"	19	"	39	"	"	"	"
Y18	18 OUT	"	"	20	"	40	"	"	"	"

1/ Group A includes pins 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20 except when connected to point B.

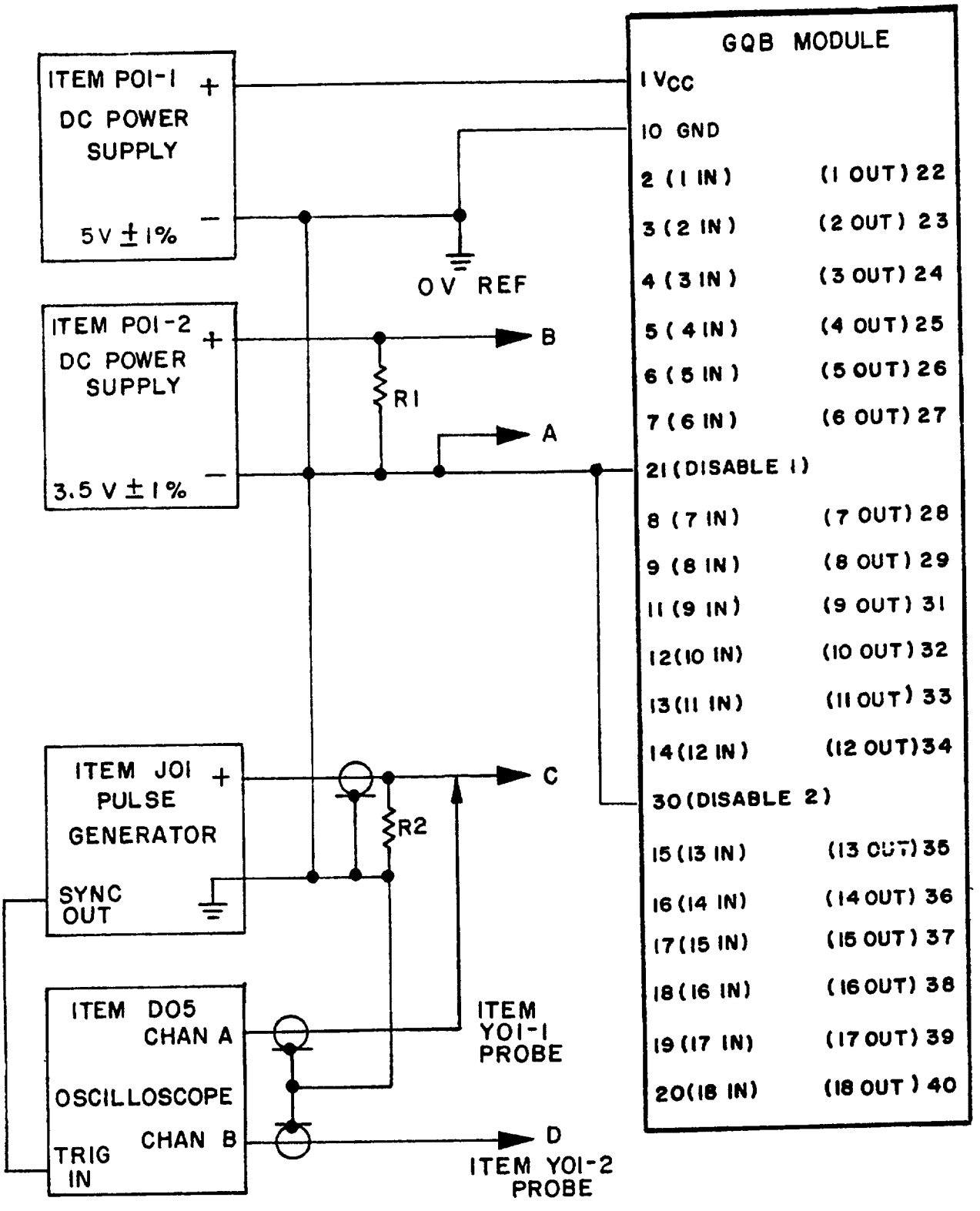
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*TABLE XXI. Output crosstalk (low-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	1 OUT	Output crosstalk, low level	2	Open	Group A 1/	22	0	1500	mV	18,19
Z2	2 OUT	"	3	"	"	23	"	"	"	"
Z3	3 OUT	"	4	"	"	24	"	"	"	"
Z4	4 OUT	"	5	"	"	25	"	"	"	"
Z5	5 OUT	"	6	"	"	26	"	"	"	"
Z6	6 OUT	"	7	"	"	27	"	"	"	"
Z7	7 OUT	"	8	"	"	28	"	"	"	"
Z8	8 OUT	"	9	"	"	29	"	"	"	"
Z9	9 OUT	"	11	"	"	31	"	"	"	"
Z10	10 OUT	"	12	"	"	32	"	"	"	"
Z11	11 OUT	"	13	"	"	33	"	"	"	"
Z12	12 OUT	"	14	"	"	34	"	"	"	"
Z13	13 OUT	"	15	"	"	35	"	"	"	"
Z14	14 OUT	"	16	"	"	36	"	"	"	"
Z15	15 OUT	"	17	"	"	37	"	"	"	"
Z16	16 OUT	"	18	"	"	38	"	"	"	"
Z17	17 OUT	"	19	"	"	39	"	"	"	"
Z18	18 OUT	"	20	"	"	40	"	"	"	"

1/ Group A includes pins 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20 except when connected to point A.

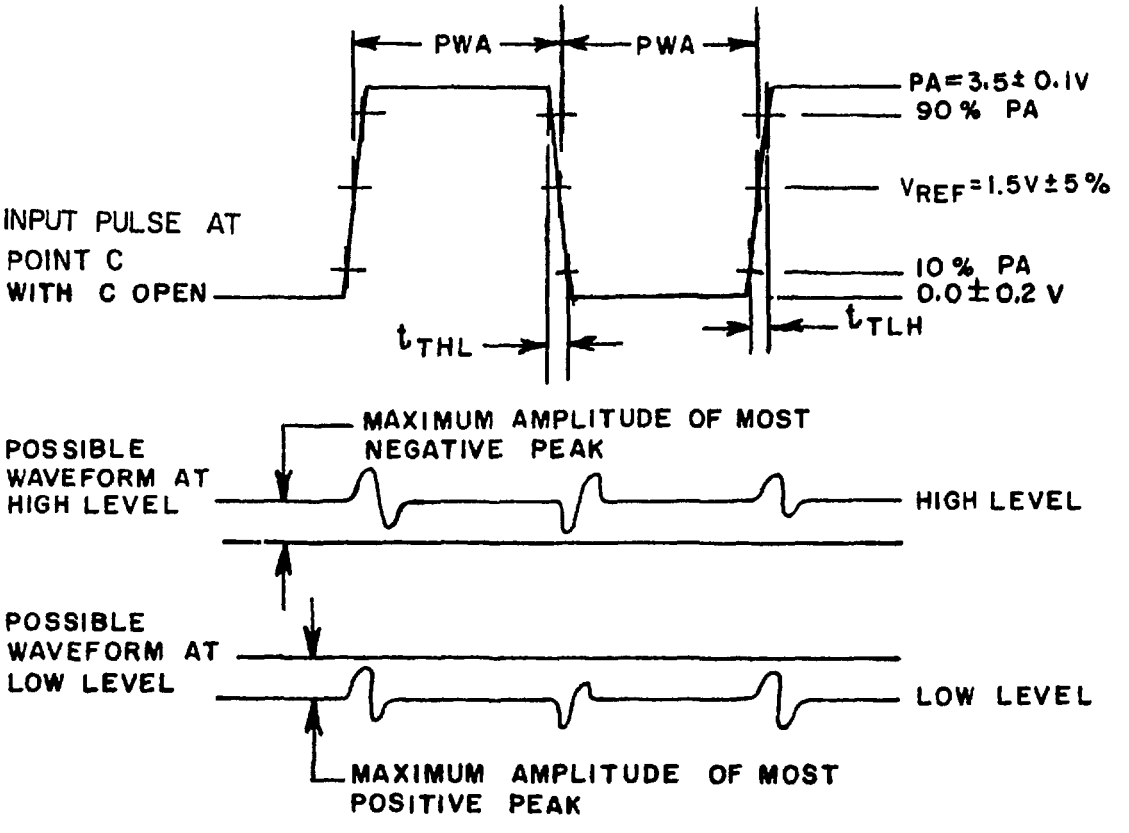
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Load all outputs with load as specified on figure 13.

FIGURE 18. Output crosstalk (high-level and low-level) test circuit for tables XX and XXI.

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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
t_{TLH} and t_{THL} = 10 ±2 ns.

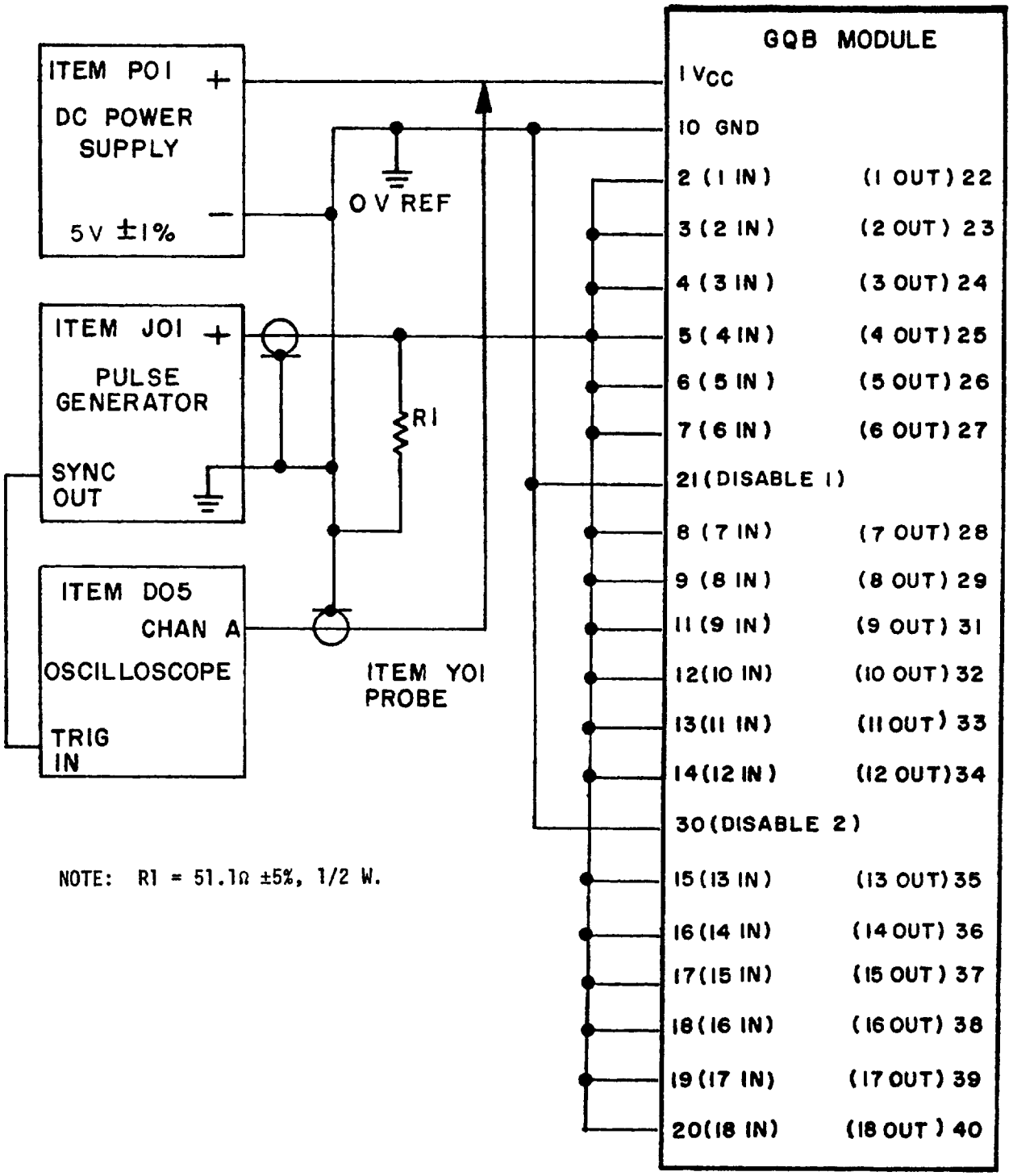
* FIGURE 19. Output crosstalk (high-level and low-level) test circuit waveforms for tables XX and XXI.

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TABLE XXII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
AA	Power supply transient amplitude	0	400	mV	20, 21

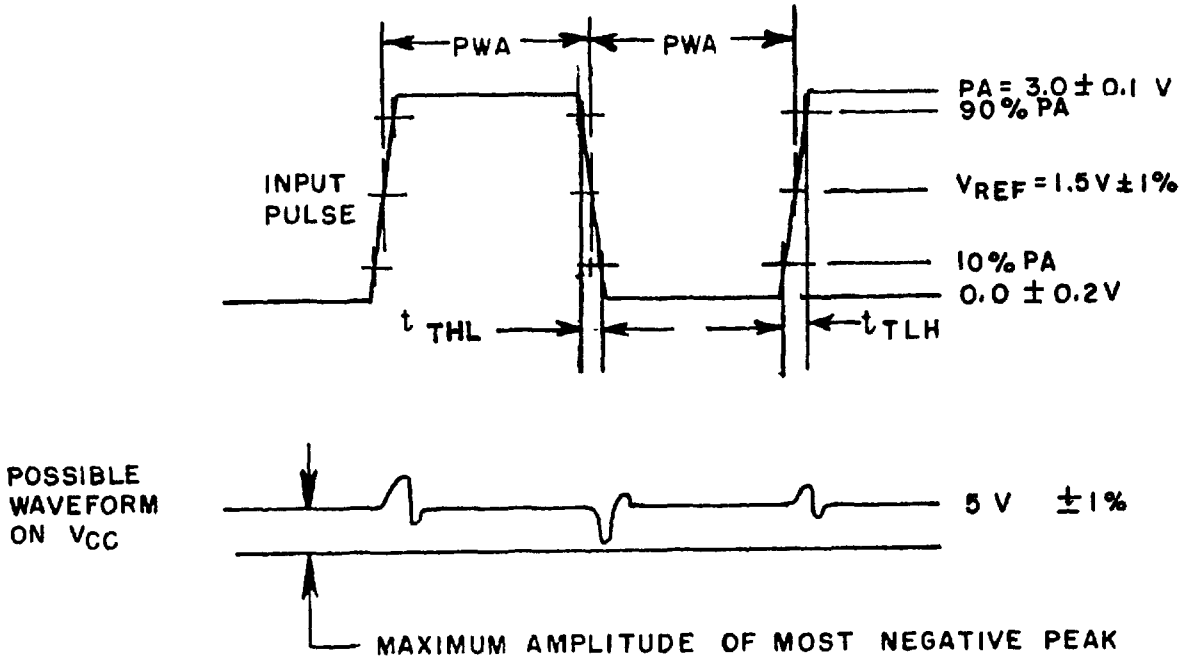
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NOTE: R1 = 51.1Ω ±5%, 1/2 W.

FIGURE 20. Power supply transient amplitude test circuit for table XXII.

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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
t_{TLH} and t_{THL} = 10 ±2 ns.

*FIGURE 21. Power supply transient amplitude test circuit waveforms for table XXII .

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TABLE XXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB	Power supply filter capacitance	Open	1	0.77	1.43	µF	22

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TABLE XXIV. *Input capacitance test.*

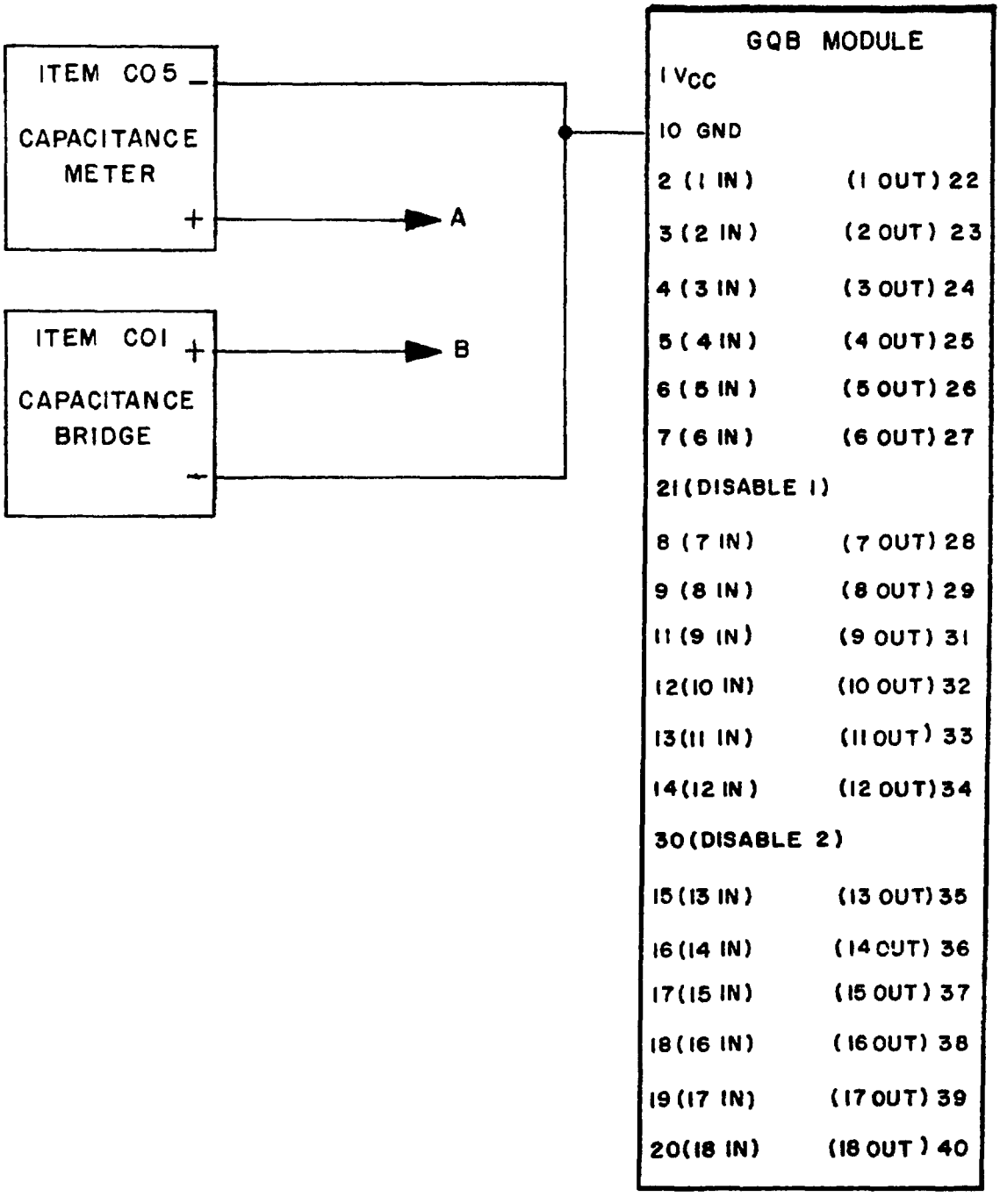
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1 IN Input capacitance	2	Open	2	15	pF	22
CC2	2 IN "	3	"	"	"	"	"
CC3	3 IN "	4	"	"	"	"	"
CC4	4 IN "	5	"	"	"	"	"
CC5	5 IN "	6	"	"	"	"	"
CC6	6 IN "	7	"	"	"	"	"
CC7	DISABLE 1 "	21	"	"	"	"	"
CC8	7 IN "	8	"	"	"	"	"
CC9	8 IN "	9	"	"	"	"	"
CC10	9 IN "	11	"	"	"	"	"
CC11	10 IN "	12	"	"	"	"	"
CC12	11 IN "	13	"	"	"	"	"
CC13	12 IN "	14	"	"	"	"	"
CC14	13 IN "	15	"	"	"	"	"
CC15	14 IN "	16	"	"	"	"	"
CC16	15 IN "	17	"	"	"	"	"
CC17	16 IN "	18	"	"	"	"	"
CC18	17 IN "	19	"	"	"	"	"
CC19	18 IN "	20	"	"	"	"	"
CC20	DISABLE 2 "	30	"	4	25	"	"

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*TABLE XXV. Output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
DD1	1 OUT	Output capacitance	22	Open	4	25	pF	22
DD2	2 OUT	"	23	"	"	"	"	"
DD3	3 OUT	"	24	"	"	"	"	"
DD4	4 OUT	"	25	"	"	"	"	"
DD5	5 OUT	"	26	"	"	"	"	"
DD6	6 OUT	"	27	"	"	"	"	"
DD7	7 OUT	"	28	"	"	"	"	"
DD8	8 OUT	"	29	"	"	"	"	"
DD9	9 OUT	"	31	"	"	"	"	"
DD10	10 OUT	"	32	"	"	"	"	"
DD11	11 OUT	"	33	"	"	"	"	"
DD12	12 OUT	"	34	"	"	"	"	"
DD13	13 OUT	"	35	"	"	"	"	"
DD14	14 OUT	"	36	"	"	"	"	"
DD15	15 OUT	"	37	"	"	"	"	"
DD16	16 OUT	"	38	"	"	"	"	"
DD17	17 OUT	"	39	"	"	"	"	"
DD18	18 OUT	"	40	"	"	"	"	"

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.
3. Item CO1 setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

* FIGURE 22. Power supply filter, input, and output capacitance test circuit for tables XXIII, XXIV, and XXV.

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	☐ OTHER *(Specify)* _____

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